



HỆ TÍNH TOÁN PHÂN BỐ

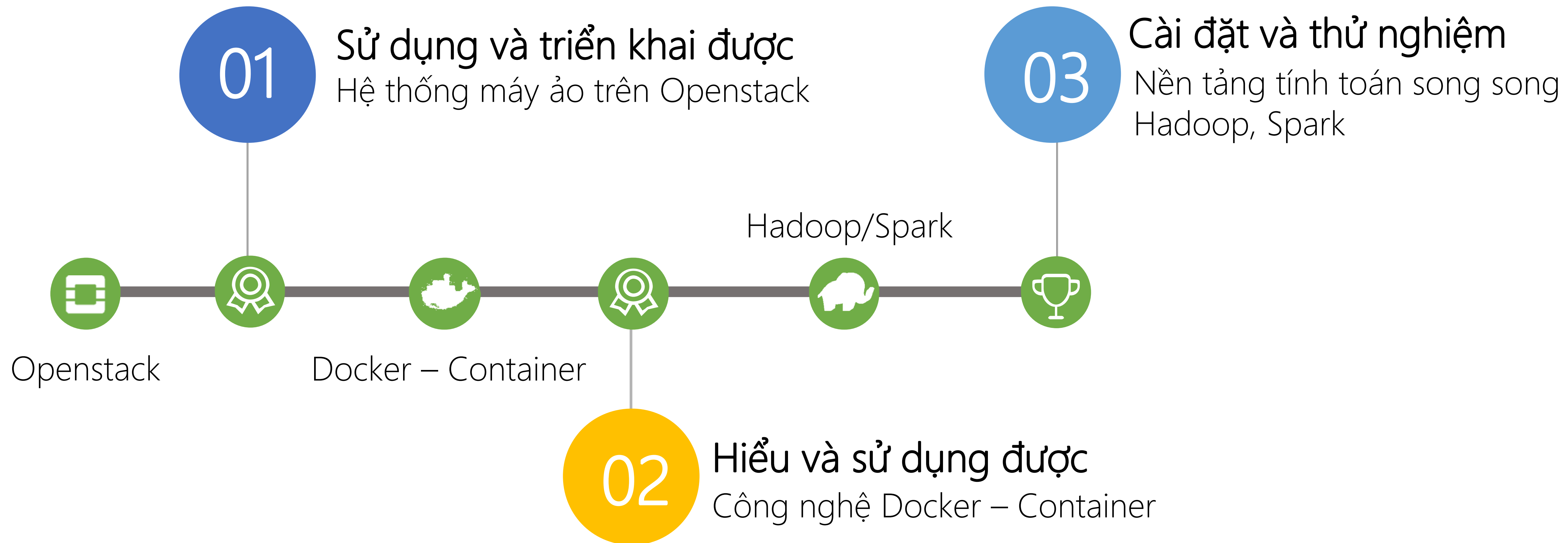
DISTRIBUTED COMPUTING

Nguyễn Văn Bảo

MỤC TIÊU

2

Chương trình thực hành



HÌNH THỨC

Thực hành và đánh giá

3



Làm Labs theo nhóm

- Kiểm tra bất kỳ nhóm/thành viên
- Vắng báo cáo phải báo trước



Deadline bài Labs

- Không thể gia hạn
- Báo cáo file
DOC/DOCX/PDF



6 bài Labs

Tính điểm Trung bình (và điểm cộng)

QUY ĐỊNH

4

NÊN



Đi học đúng giờ



Tuân thủ deadline



Đọc kỹ hướng dẫn, yêu cầu



Sử dụng đúng timeslot

4

KHÔNG

X

“Gánh team”

X

Đổi nhóm

X

Sao chép

X

“Đổ thừa” tại, bị



Ha Huy Hoang

LAB 1

OPENSTACK – WEB UI

NỘI DUNG

có những gì?

6



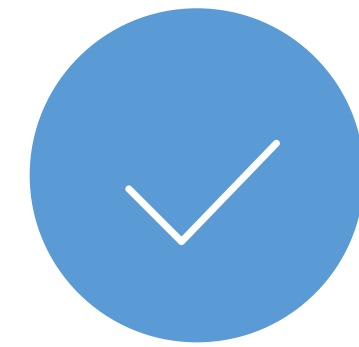
Mục tiêu bài Lab



Giới thiệu Openstack



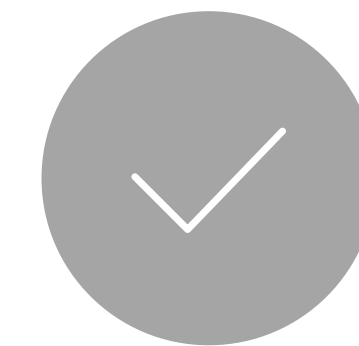
Các phương thức kết nối,
điều khiển Openstack



Giới thiệu Openstack Web UI



Demo các thao tác cơ bản



Yêu cầu thực hành

HỌC GÌ VÀ LÀM ĐƯỢC GÌ

hôm nay?

7

Tạo Network/
Router



Tạo Keypair



Tạo Snapshot



Manage
Firewall Rules



Sử dụng
Floating IP



Sử dụng
InitScripts



Triển khai, cấu hình

Hệ thống máy ảo theo
kiến trúc mạng cho trước

OPENSTACK

là gì?

8

A

dự án mã nguồn mở
dùng để triển khai
cloud computing

B

bao gồm nhiều thành
phần do cộng đồng
phát triển

A

C

B

D

C

Rackspace & NASA
công bố tháng 10/2010

D

Công bố phiên bản
mới mỗi 6 tháng

OPENSTACK

“Open source software for building private and public clouds” - <https://www.openstack.org/>

CÁC PHƯƠNG THỨC KẾT NỐI, ĐIỀU KHIỂN

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Hệ thống Openstack - UIT

Web UI



Sử dụng giao diện web
<http://mitaka.uit.edu.vn>

CLI



Sử dụng Python
OpenstackClient

Rest APIs

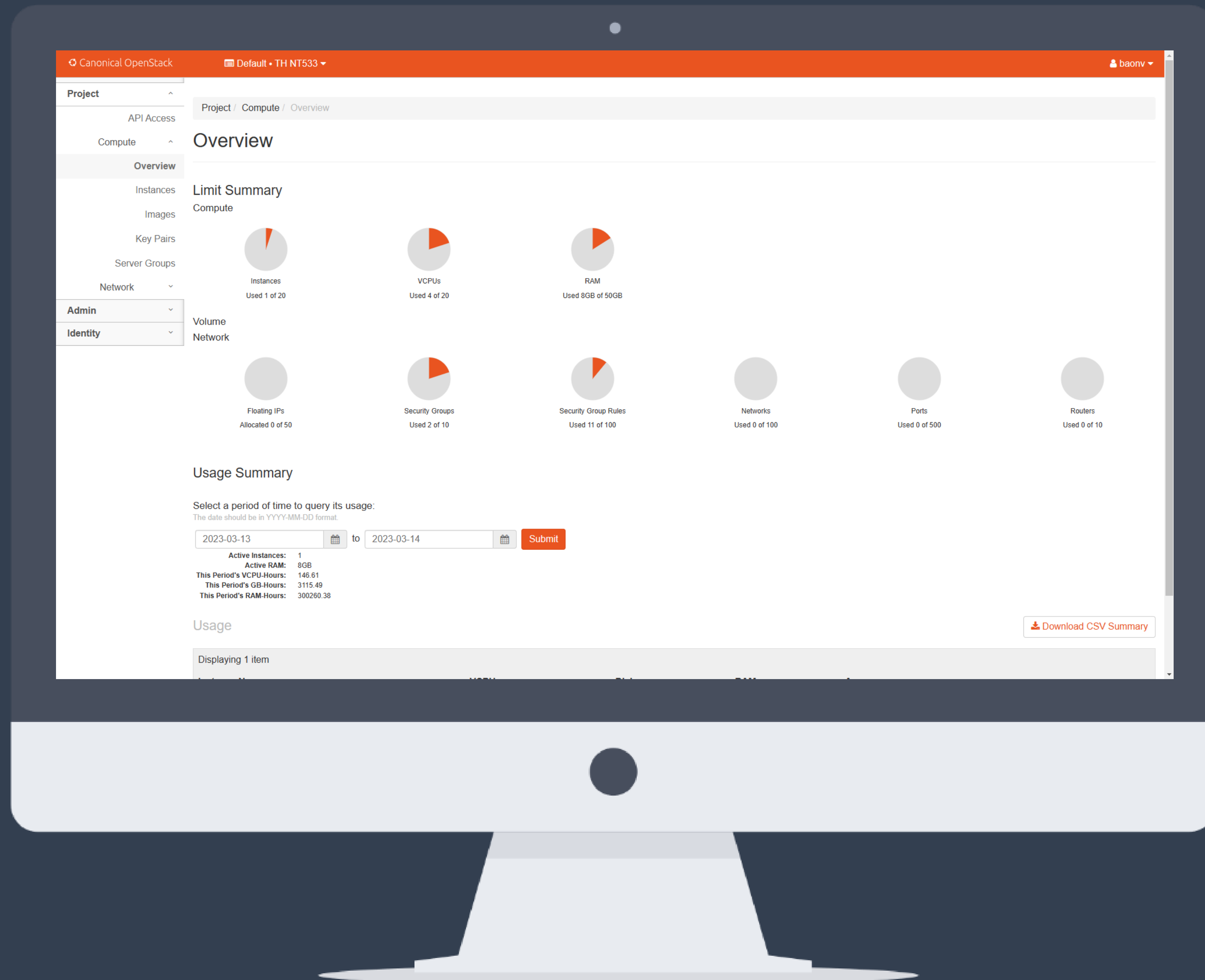


Để phát triển ứng dụng trên web/apps.
Có thể test bằng Advanced REST Client
trên Chrome

GIỚI THIỆU GIAO DIỆN OPENSTACK WEB UI

Demo trực tiếp tại lớp

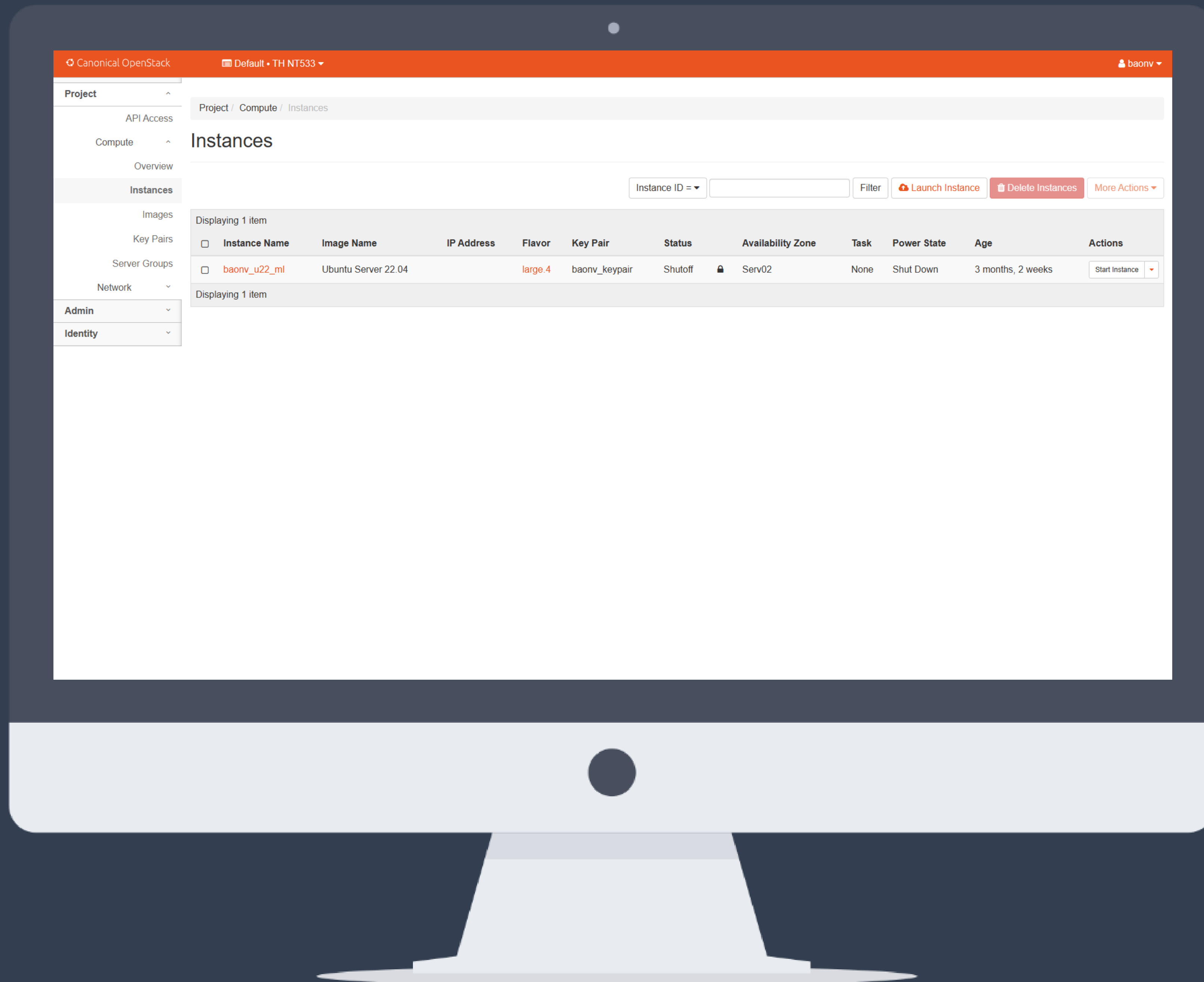




OVERVIEW

Thống kê tổng quan
lượng tài nguyên đang
được sử dụng

Demo trực tiếp tại lớp



INSTANCE

Các thao tác trên instance (máy ảo):

- Launch/Delete
- Gắn Floating IP
- Reboot/Shutdown
- Lock/Resize
- ...

Demo trực tiếp tại lớp

Canonical OpenStack

Default • TH NT533

baonv

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Network

Admin

Identity

Project / Compute / Images

Images

Click here for filters or full text search.

Create Image

Delete Images

Displaying 11 items

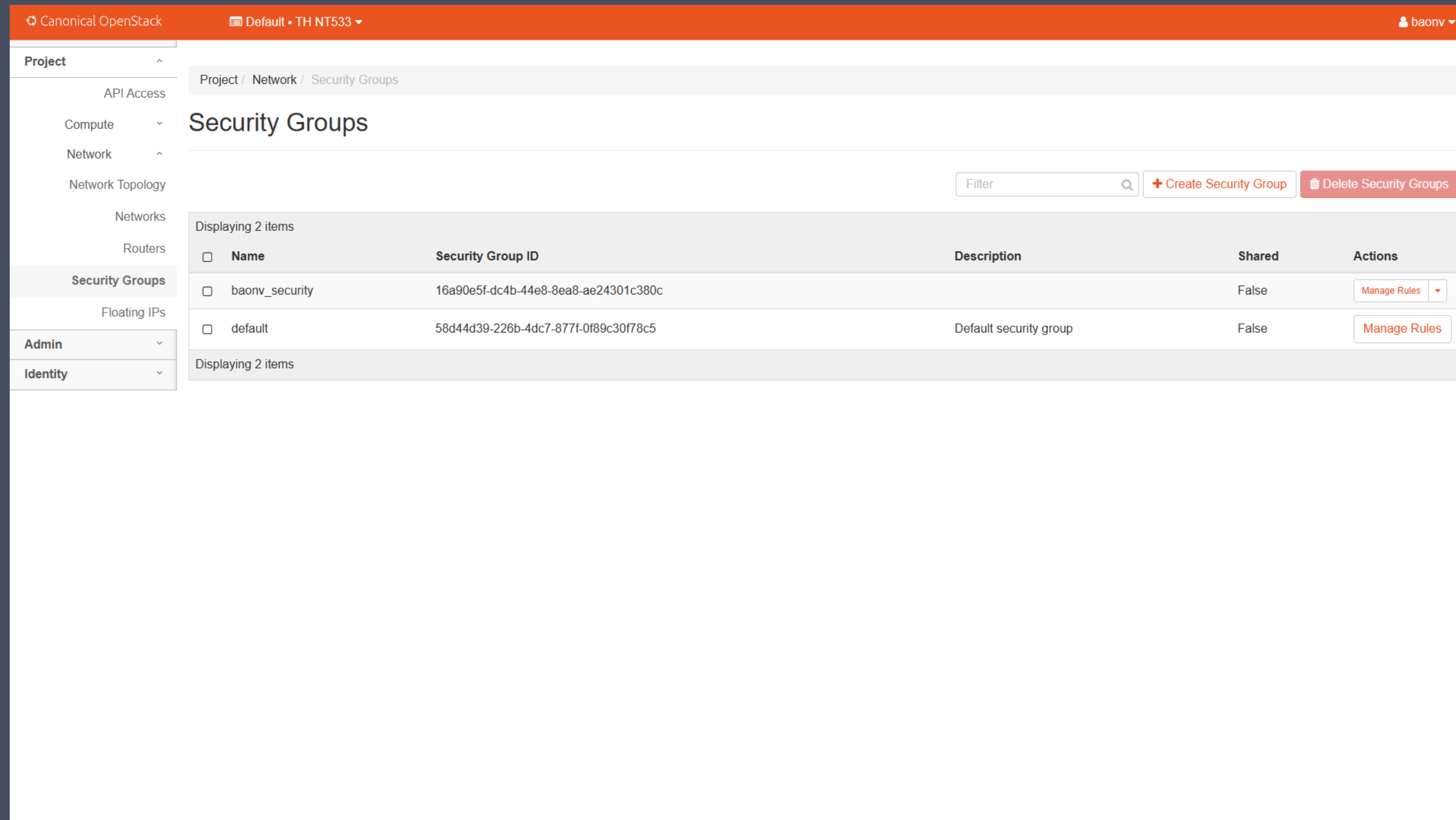
	Owner	Name ^	Type	Status	Visibility	Protected	Disk Format	Size	
☐	admin	CentOS 6	Image	Active	Public	Yes	QCOW2	769.38 MB	Launch
☐	admin	CentOS 7	Image	Active	Public	Yes	QCOW2	819.00 MB	Launch
☐	admin	CentOS 8 Stream	Image	Active	Public	Yes	QCOW2	1.45 GB	Launch
☐	admin	CentOS 9 Stream	Image	Active	Public	Yes	QCOW2	891.38 MB	Launch
☐	admin	Cirros-0.5.1	Image	Active	Public	No	QCOW2	15.58 MB	Launch
☐	admin	EVE-22.06.13	Image	Active	Image from Other Project - Non-Public	No	QCOW2	635.88 MB	Launch
☐	admin	Kali-2022	Image	Active	Public	No	QCOW2	80.00 GB	Launch
☐	admin	Ubuntu Server 16.04	Image	Active	Public	Yes	QCOW2	300.75 MB	Launch
☐	admin	Ubuntu Server 18.04	Image	Active	Public	Yes	QCOW2	371.63 MB	Launch
☐	admin	Ubuntu Server 20.04	Image	Active	Public	Yes	QCOW2	575.44 MB	Launch
☐	admin	Ubuntu Server 22.04	Image	Active	Public	No	QCOW2	631.00 MB	Launch

Displaying 11 items

IMAGES

- Quản lý Images:
- Create/Delete
 - Launch instance từ image

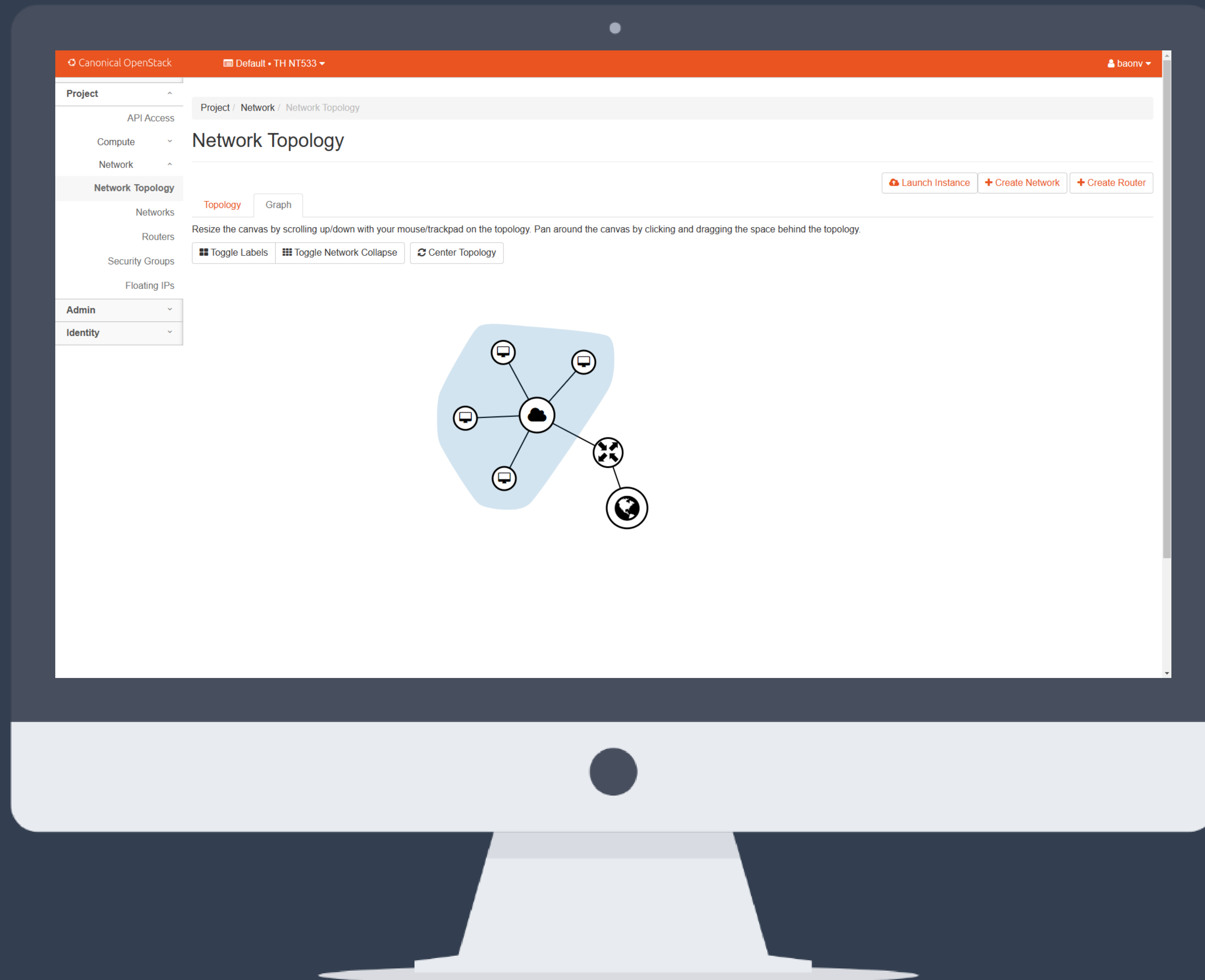
Demo trực tiếp tại lớp



ACCESS & SECURITY

- Quản lý rules firewall
- Quản lý keypair
- Quản lý Floating Ips
- Thông tin API Access

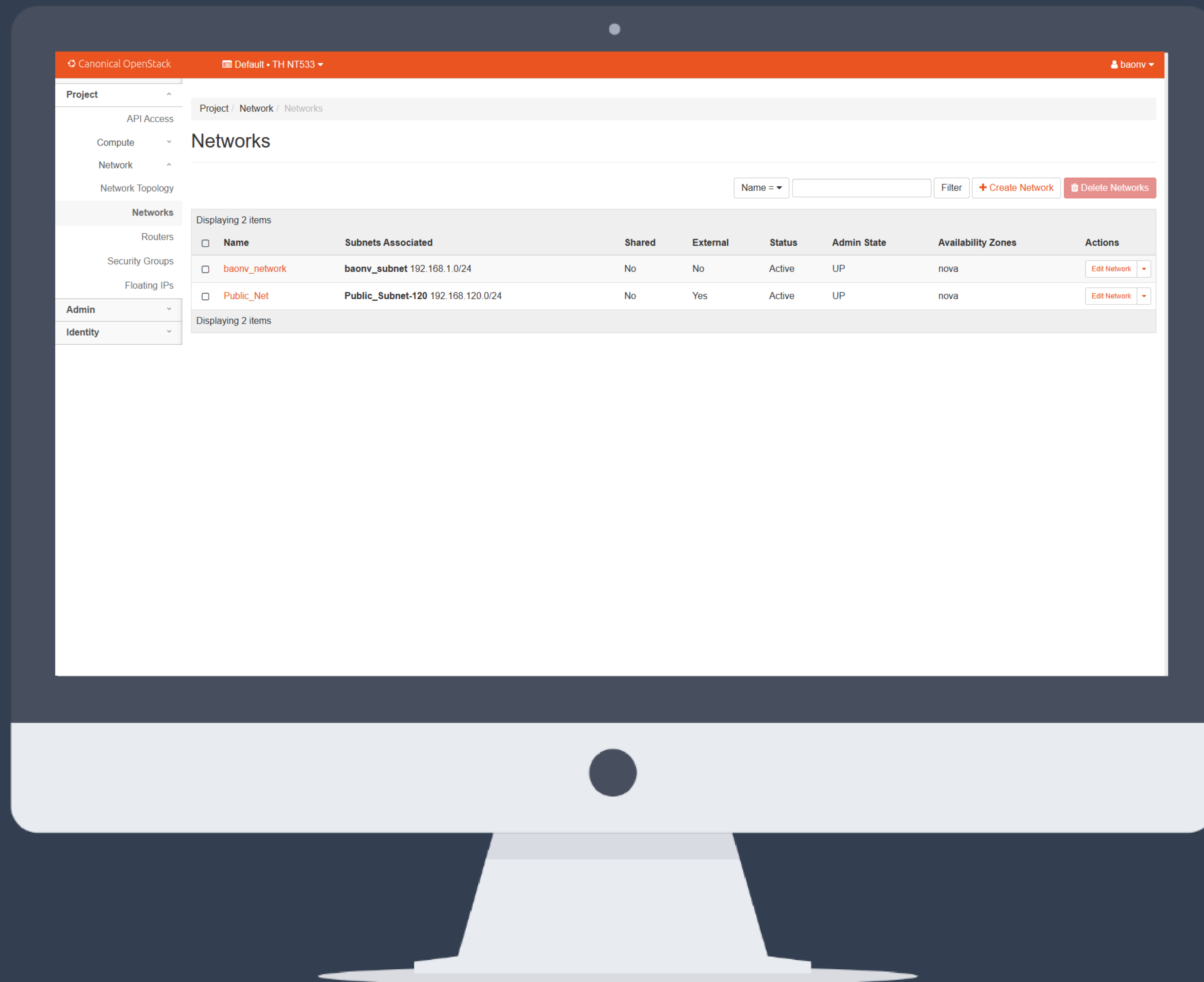
Demo trực tiếp tại lớp



NETWORK TOPOLOGY

Mô hình mạng kết nối các máy ảo

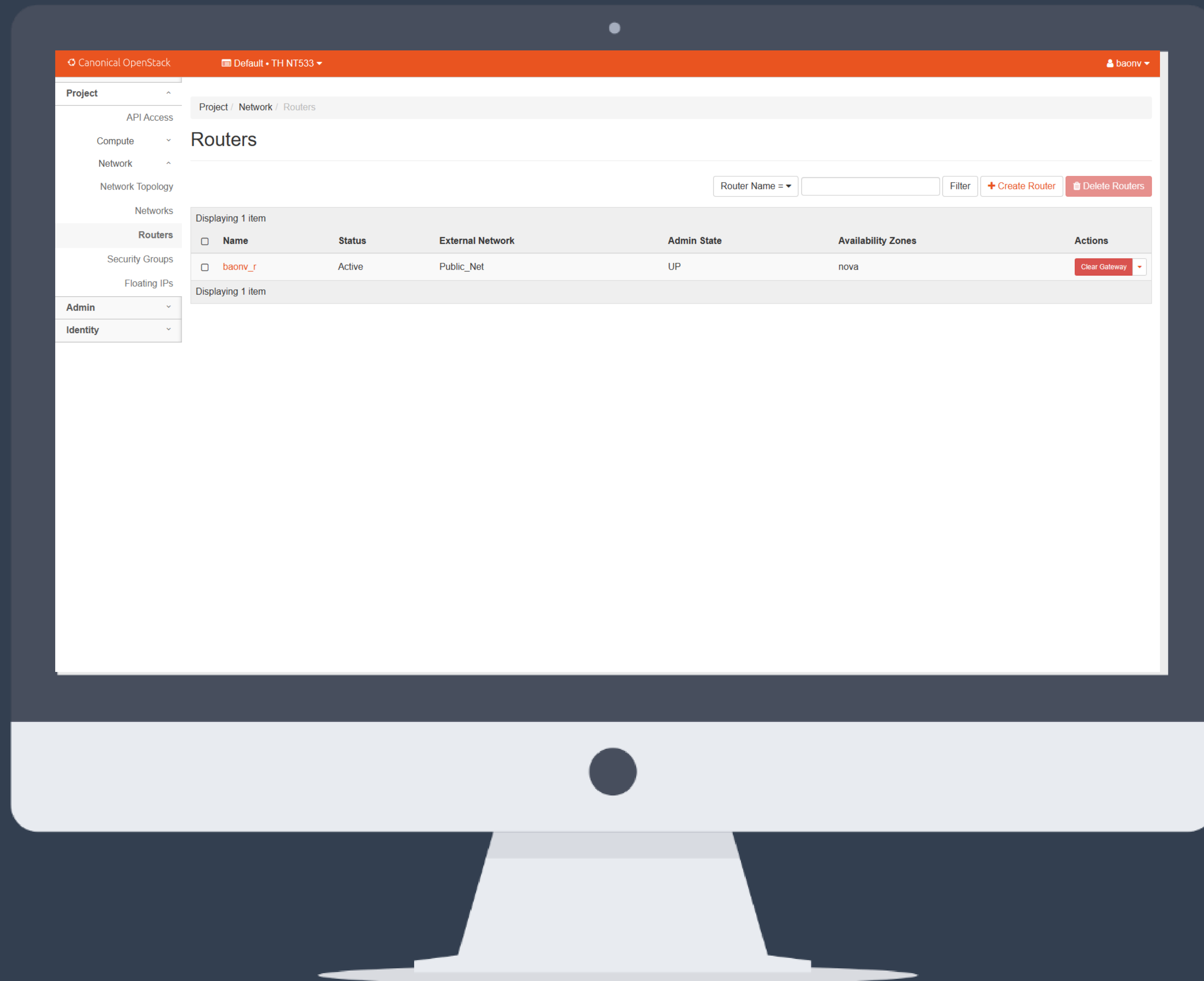
Demo trực tiếp tại lớp



NETWORKS

Các thao tác với Networks:
Create/Edit/Delete

Demo trực tiếp tại lớp



ROUTERS

Create/Delete/Edit Routers

Demo trực tiếp tại lớp

GIỚI THIỆU CÁC THAO TÁC CƠ BẢN



Demo trực tiếp tại lớp

Canonical OpenStack

Default • TH NT533

Project

API Access

Compute

Network

Network Topology

Networks

Routers

Security Groups

Floating IPs

Admin

Identity

Project / Network

Network

Displaying 2

☐ Name

☐ baon

☐ Publi

Displaying 2

Create Network

Network

Subnet

Subnet Details

Network Name

nhom01_net

☒ Enable Admin State

☐ Shared

☒ Create Subnet

Availability Zone Hints

nova

MTU

Create a new network. In addition, a subnet associated with the network can be created in the following steps of this wizard.

+ Create Network

Availability Zone

nova

nova

Cancel

« Back

Next »

Create Network

Network

Subnet

Subnet Details

Subnet Name

nhom01_subnet1

Network Address ?

192.168.1.0/24

IP Version

IPv4

Gateway IP ?

☐ Disable Gateway

Creates a subnet associated with the network. You need to enter a valid "Network Address" and "Gateway IP". If you did not enter the "Gateway IP", the first value of a network will be assigned by default. If you do not want gateway please check the "Disable Gateway" checkbox. Advanced configuration is available by clicking on the "Subnet Details" tab.

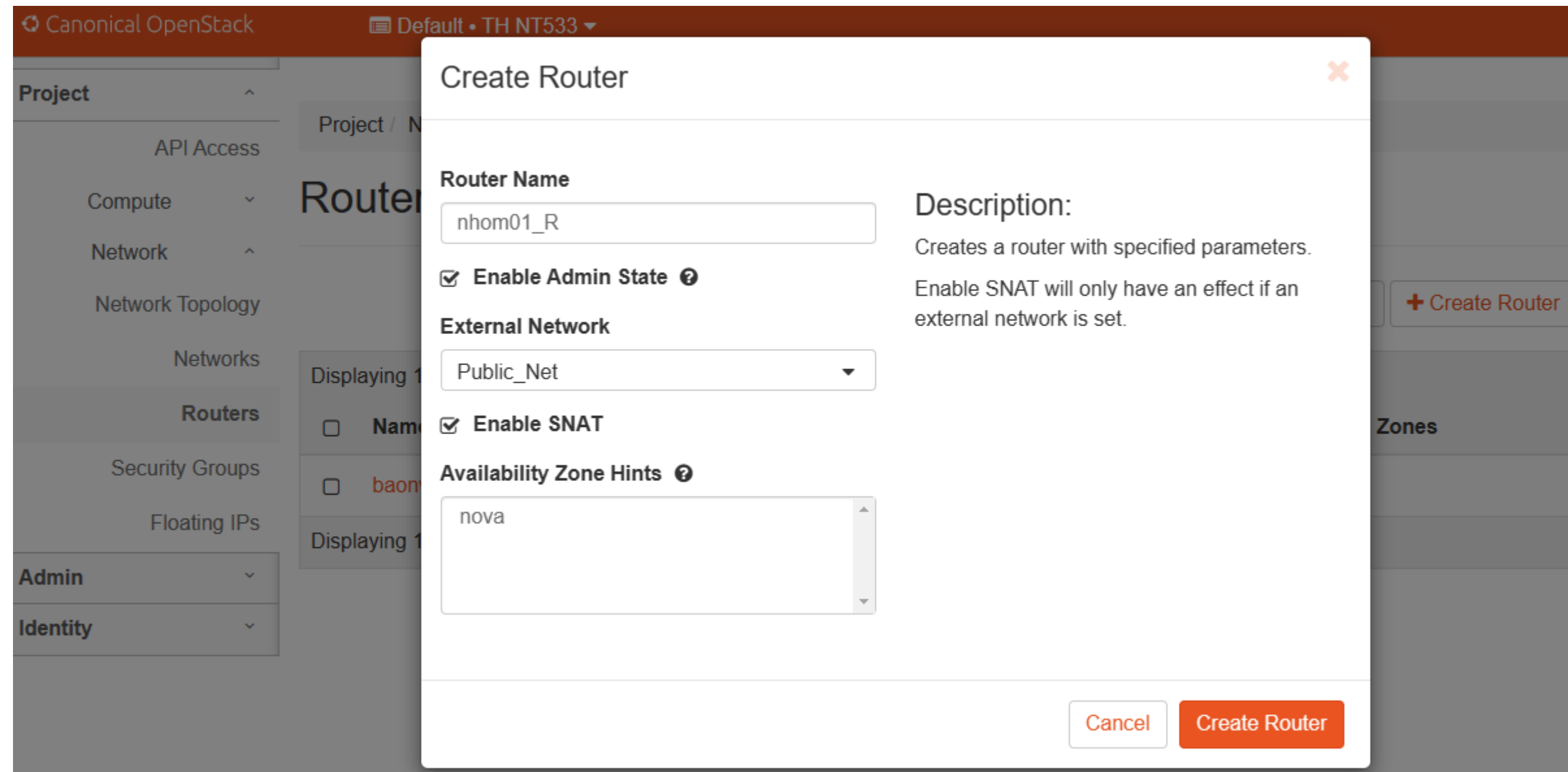
Cancel

« Back

Next »

Demo trực tiếp tại lớp

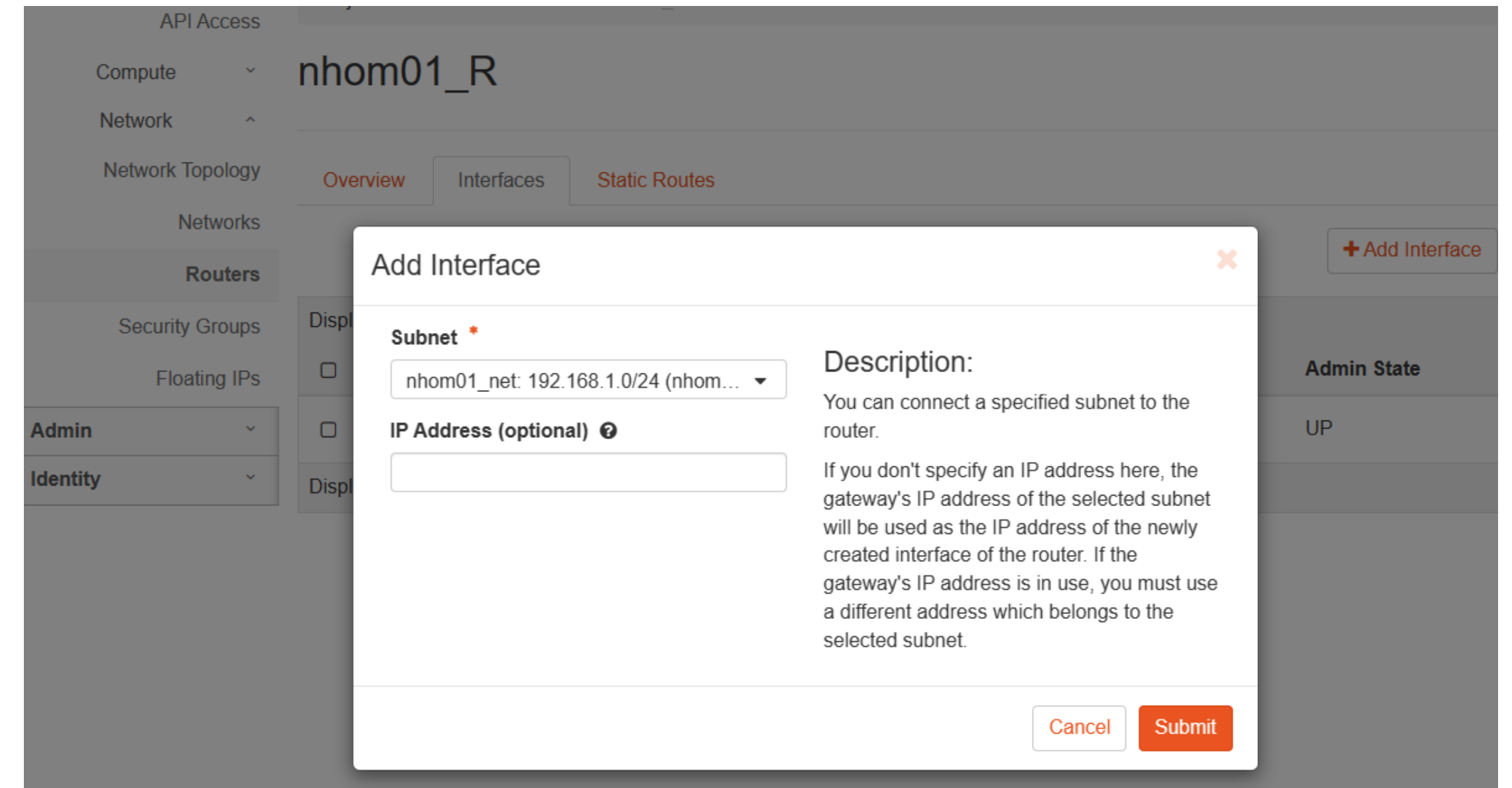
TẠO NETWORK



The screenshot shows the OpenStack dashboard with the 'Create Router' dialog box open. The dialog box has a title bar with a close button. It contains the following fields and options:

- Router Name:** A text input field containing 'nhom01_R'.
- Description:** A text area with the text 'Creates a router with specified parameters. Enable SNAT will only have an effect if an external network is set.'
- Enable Admin State:** A checked checkbox with a help icon.
- External Network:** A dropdown menu showing 'Public_Net'.
- Enable SNAT:** A checked checkbox.
- Availability Zone Hints:** A text input field containing 'nova'.

At the bottom of the dialog box are two buttons: 'Cancel' and 'Create Router'.



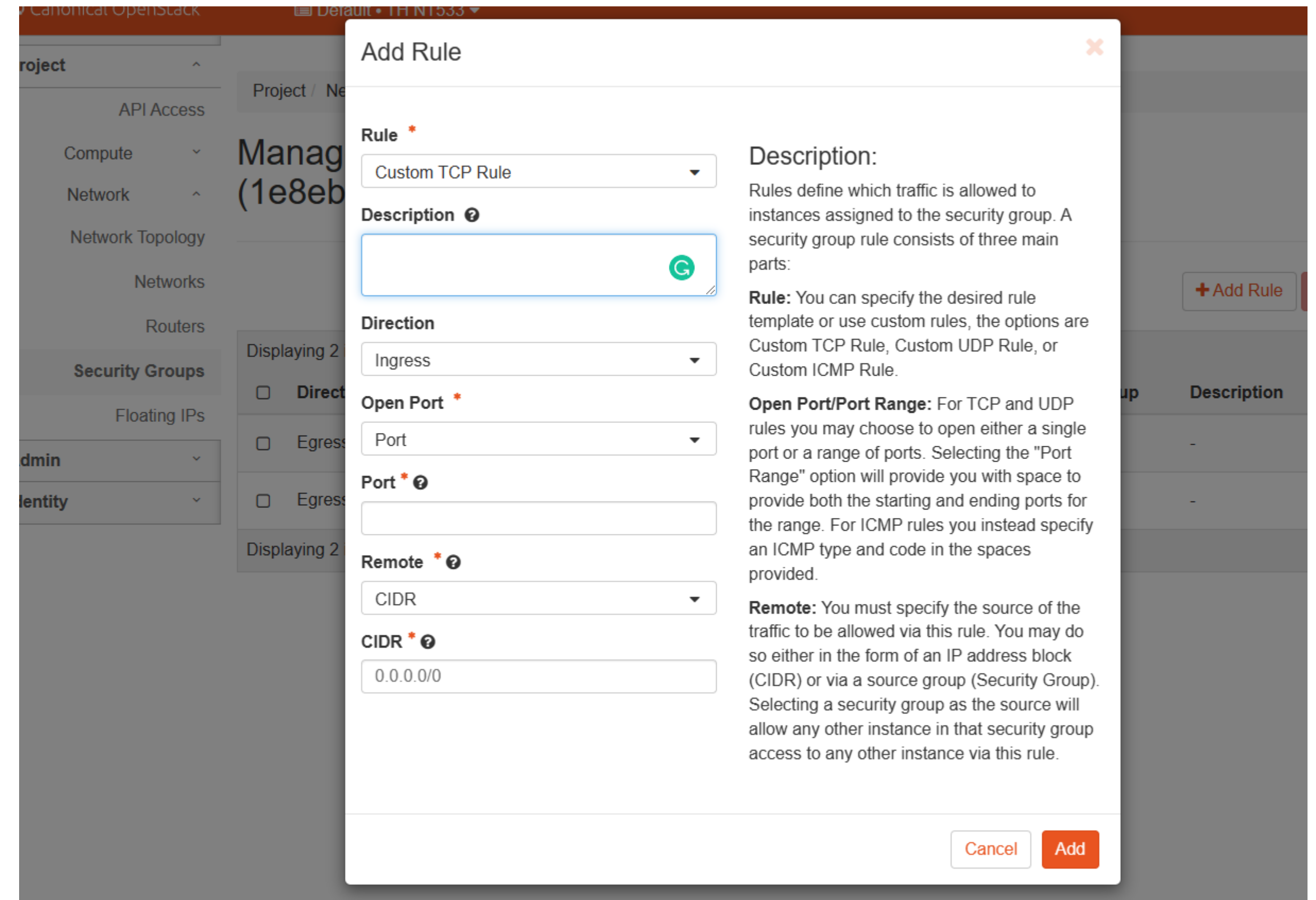
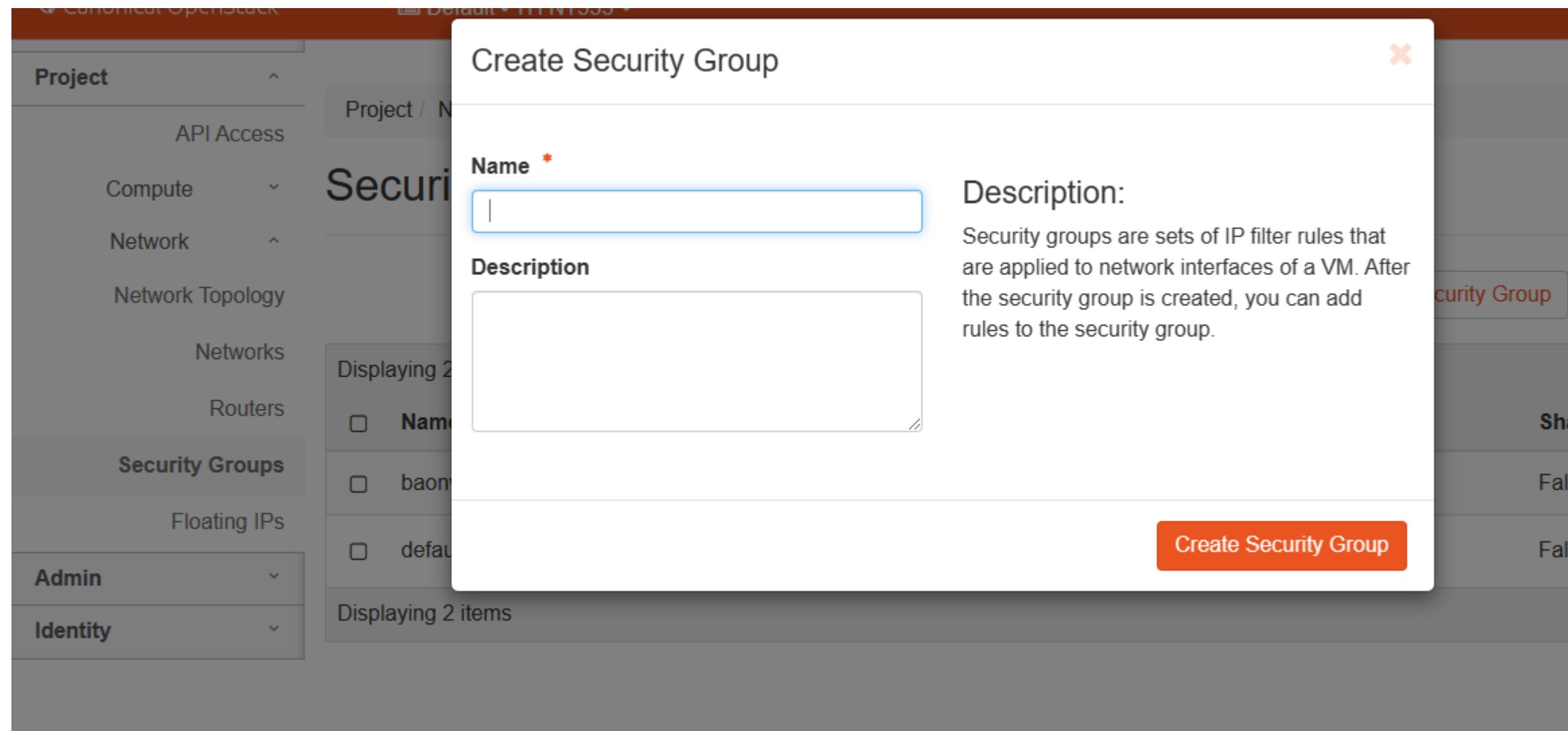
The screenshot shows the OpenStack dashboard with the 'Add Interface' dialog box open for the router 'nhom01_R'. The dialog box has a title bar with a close button. It contains the following fields and options:

- Subnet:** A dropdown menu showing 'nhom01_net: 192.168.1.0/24 (nhom...'.
- IP Address (optional):** A text input field.
- Description:** A text area with the text 'You can connect a specified subnet to the router. If you don't specify an IP address here, the gateway's IP address of the selected subnet will be used as the IP address of the newly created interface of the router. If the gateway's IP address is in use, you must use a different address which belongs to the selected subnet.'

At the bottom of the dialog box are two buttons: 'Cancel' and 'Submit'.

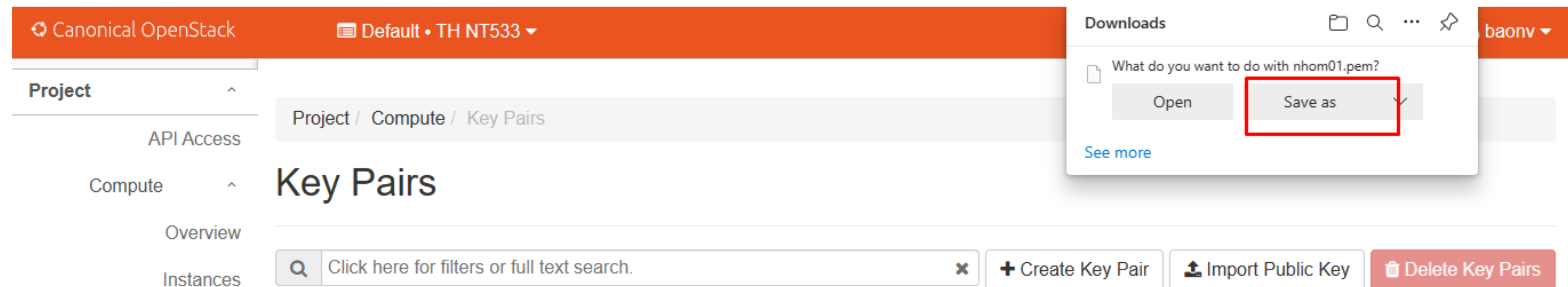
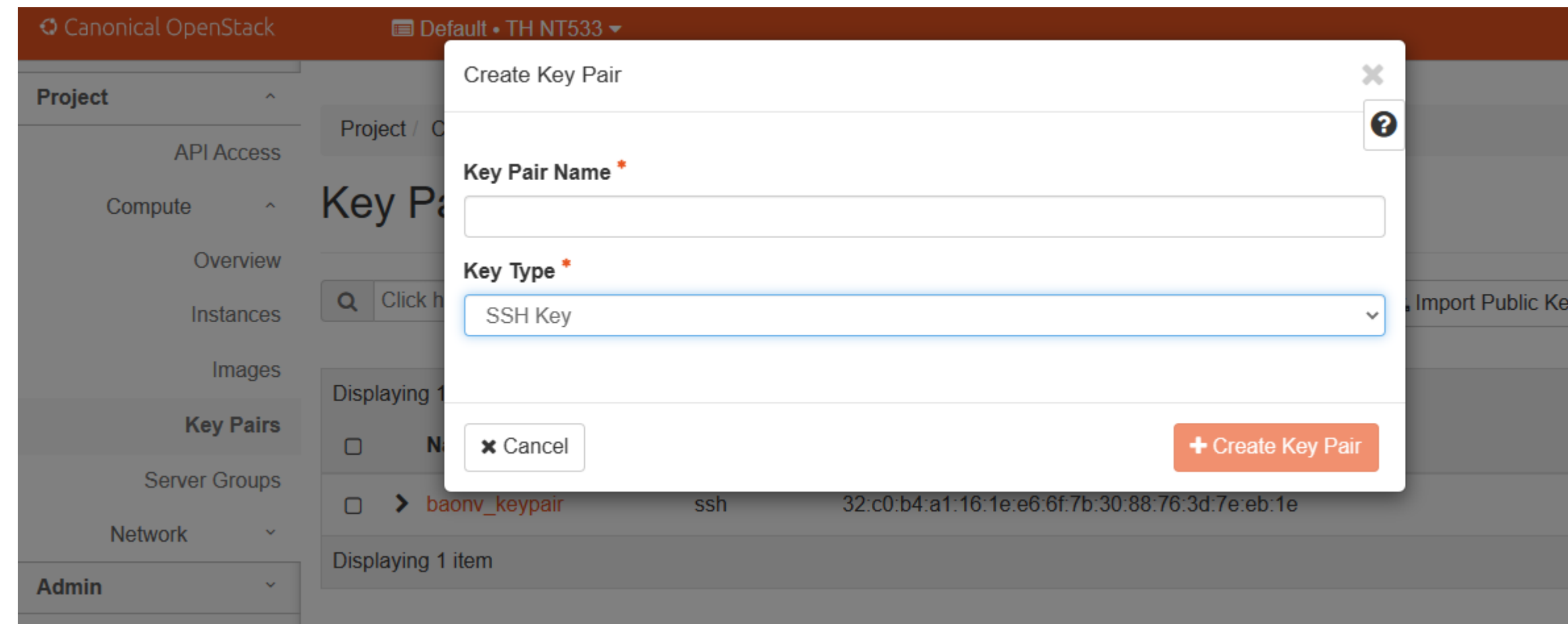
Demo trực tiếp tại lớp

TẠO ROUTER



Demo trực tiếp tại lớp

THÊM RULES FIREWALL



Demo trực tiếp tại lớp

TẠO KEYPAIR

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Metadata

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Instance Name

nhom01_pc0

Availability Zone

nova

Count

1

Total Instances (32 Max)

81%

25

Current Usage

1

Added

6

Remaining

Cancel

Back

Next

Launch Instance

Instance source is the template used to create an instance. You can use a snapshot of an existing instance, an image, or a volume (if enabled). You can also choose to use persistent storage by creating a new volume.

Select Boot Source

Image

Create New Volume

Yes

No

Allocated

NAME	UPDATED	SIZE	TYPE	VISIBILITY	
centos7	5/10/17 7:51 AM	958.44 MB	QCOW2	Public	-

Available

Click here for filters.

NAME	UPDATED	SIZE	TYPE	VISIBILITY	
debian8	5/10/17 7:53 AM	447.13 MB	QCOW2	Public	+

Flavors manage the sizing for the compute, memory and storage capacity of the instance.

Allocated

NAME	VCPUS	RAM	TOTAL DISK	PUBLIC	
m1.medium	2	4 GB	40 GB	Yes	-

Available

Click here for filters.

NAME	VCPUS	RAM	TOTAL DISK	PUBLIC	
m1.micro	1	64 MB	0 GB	Yes	+
m1.tiny	1	512 MB	1 GB	Yes	+
m1.small	1	2 GB	20 GB	Yes	+
m1.large	4	8 GB	80 GB	Yes	+

Networks provide the communication channels for instances in the cloud.

Allocated

Select networks from those listed below.

	NETWORK	SHARED	ADMIN STATE	STATUS	
1	local_all	No	Up	Active	-
2	net1	No	Up	Active	-
3	net2	No	Up	Active	-
4	net3	No	Up	Active	-

Available

Select at least one network

Click here for filters.

NETWORK	SHARED	ADMIN STATE	STATUS
No available items			

Back

Next

Launch Instance

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Metadata

A key pair allows you to SSH into your newly created instance. You may select an existing key pair, import a key pair, or generate a new key pair.

A key pair named 'nhom01' was successfully created. This key pair should automatically download. If not, you can manually download this keypair at the following link:
nhom01
Note: you will not be able to download this later.

Create Key Pair

Import Key Pair

Allocated

NAME	
nhom01	-

Available

Filter

NAME	FINGERPRINT	
nhan	fb:4f:59:31:49:7f:a5:9b:53:b0:d3:f3:1d:ae:0a:88	+

You can customize your instance after it has launched using the options available here. "Customization Script" is analogous to "User Data" in other systems.

Customization Script

Script size: 0 bytes

Load script from a file

Choose File

No file chosen

Disk Partition

Automatic

Configuration Drive

Back

Next

Launch Instance

Demo trực tiếp tại lớp

TẠO INSTANCE

Displaying 4 items

☐	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status		Availability Zone	Task	Power State	Age	Actions
☐	tt-3	Ubuntu Server 22.04	192.168.1.226	small.1	baonv_keypair	Active		Serv03	None	Running	12 minutes	Create Snapshot
☐	tt-2	Ubuntu Server 22.04	192.168.1.246	small.1	baonv_keypair	Active		Serv03	None	Running	12 minutes	Associate Floating IP
☐	tt-1	Ubuntu Server 22.04	192.168.1.49	small.1	baonv_keypair	Active		Serv02	None	Running	12 minutes	Attach Interface
☐	baonv_u22_ml	Ubuntu Server 22.04	192.168.1.35	large.4	baonv_keypair	Shutoff		Serv02	None	Shut Down	3 months, 2 weeks	Detach Interface

Displaying 4 items

- Edit Instance
- Update Metadata
- Edit Security Groups
- Edit Port Security Groups
- Console
- View Log
- Rescue Instance
- Reboot Instance

Demo trực tiếp tại lớp

GẮN FLOATING IP

KẾT NỐI SSH



vào máy ảo

Truy cập mạng
Private

Tạo Private Key

Kết nối SSH

KẾT NỐI SSH

vào máy ảo

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Truy cập mạng
Private

Tạo Private Key

Kết nối SSH

VPN

Dùng OpenVPN

Sử dụng Account sinh viên

Có thể bị lỗi với ISP FPT

Hướng dẫn: <https://phongdl.uit.edu.vn/su-dung-openvpn>

KẾT NỐI SSH

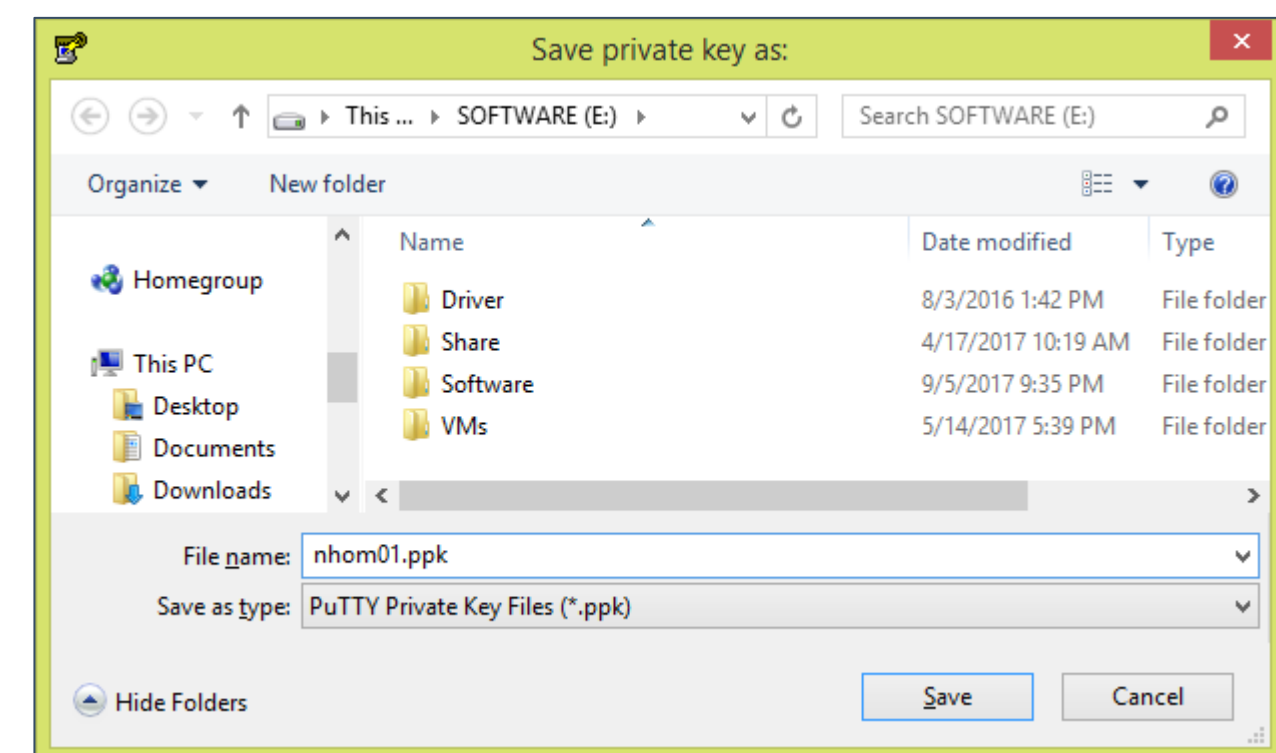
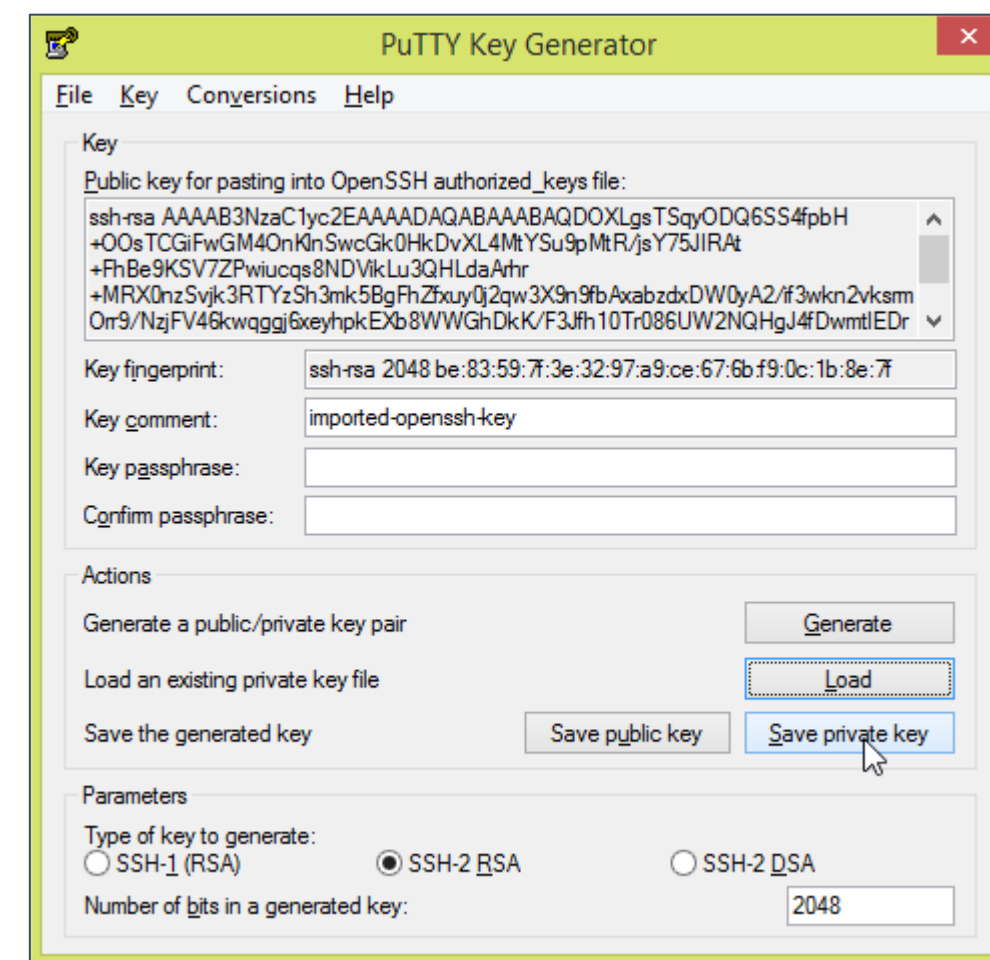
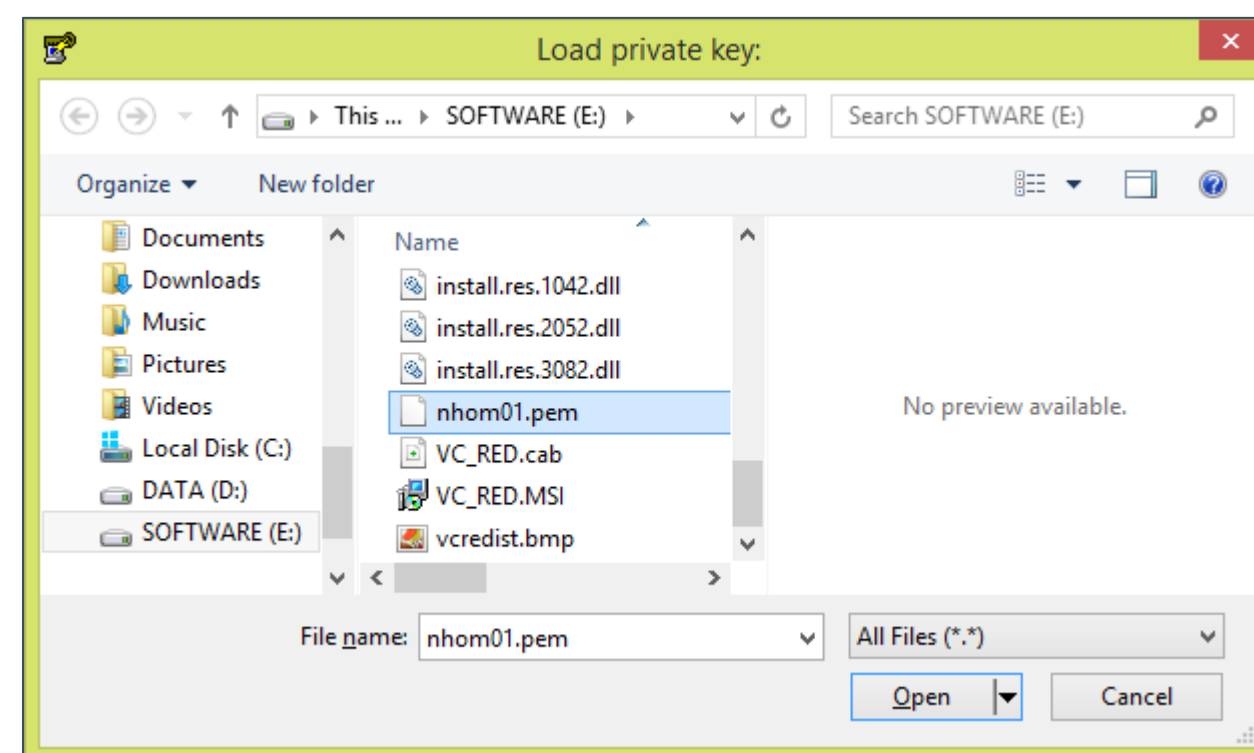
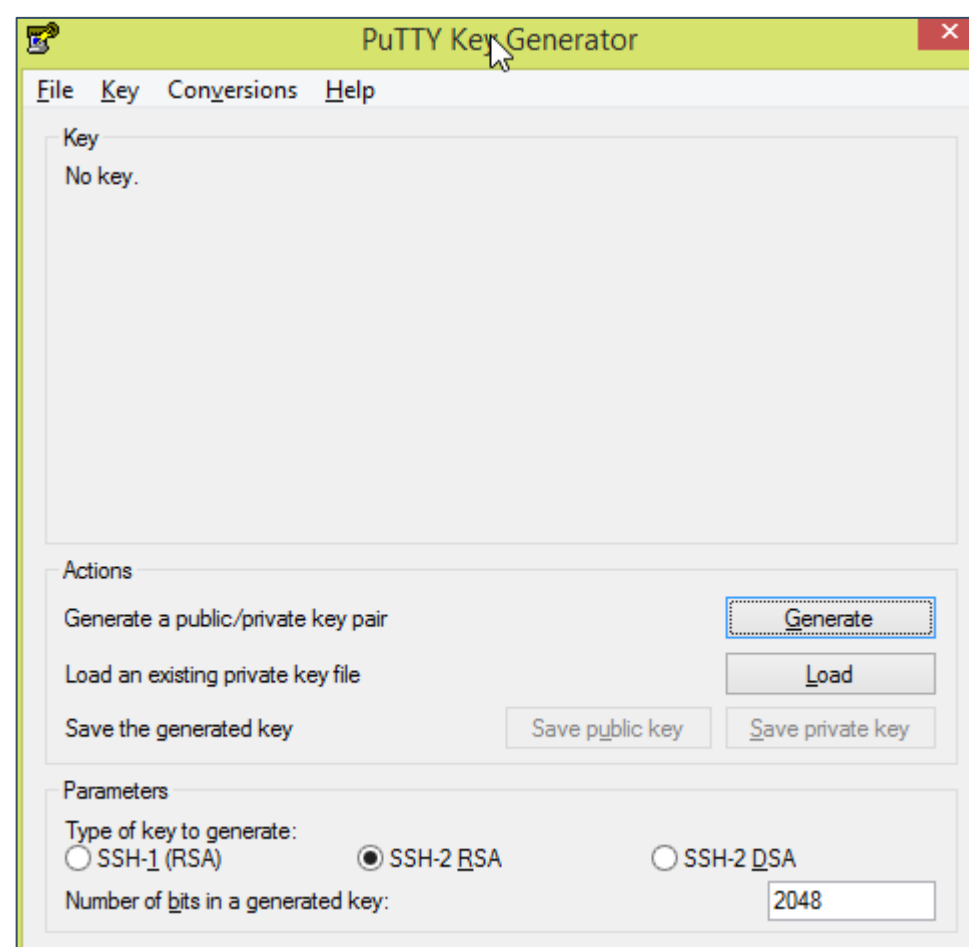
vào máy ảo

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Truy cập mạng
Private

Tạo Private Key

Kết nối SSH



KẾT NỐI SSH

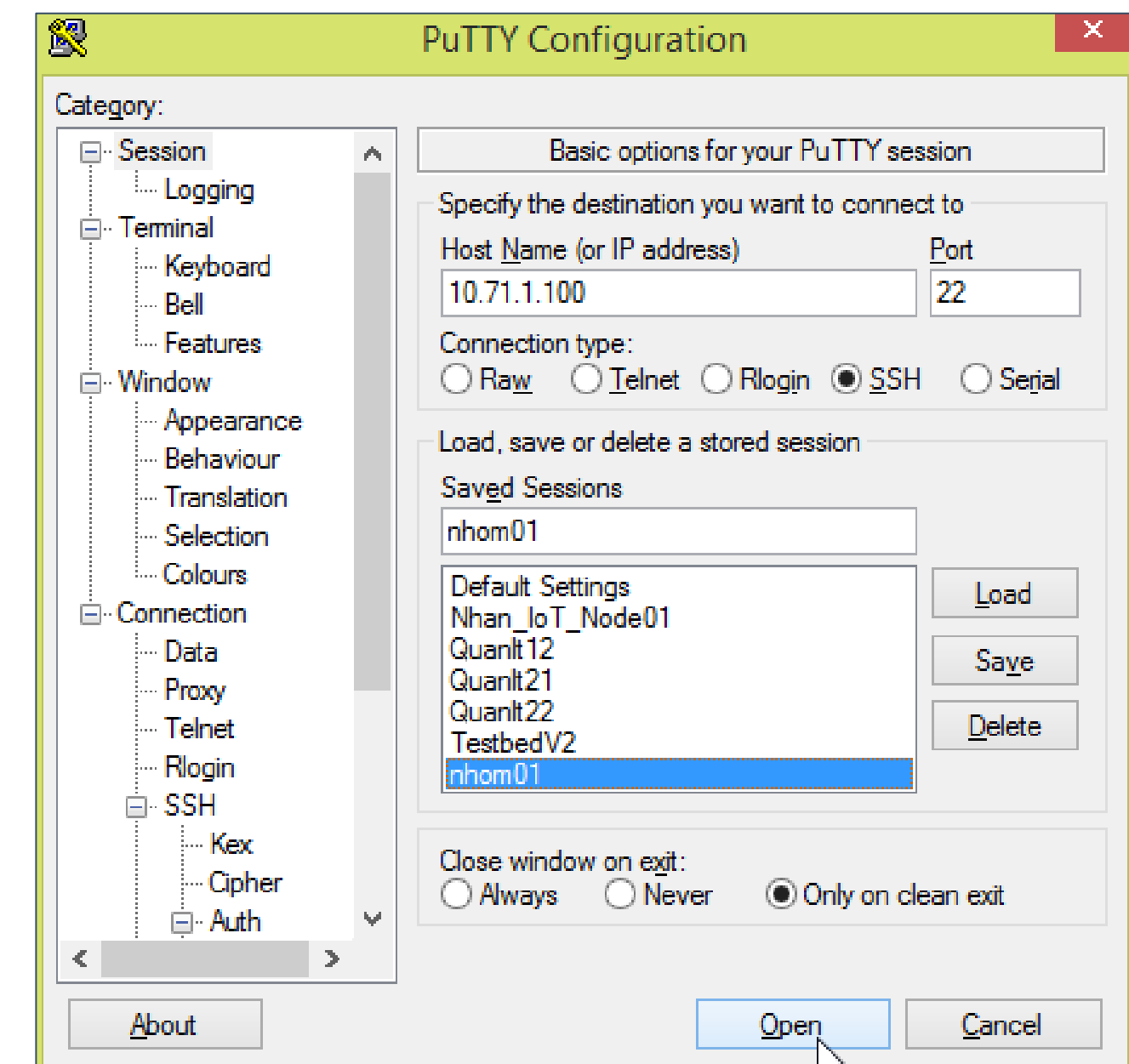
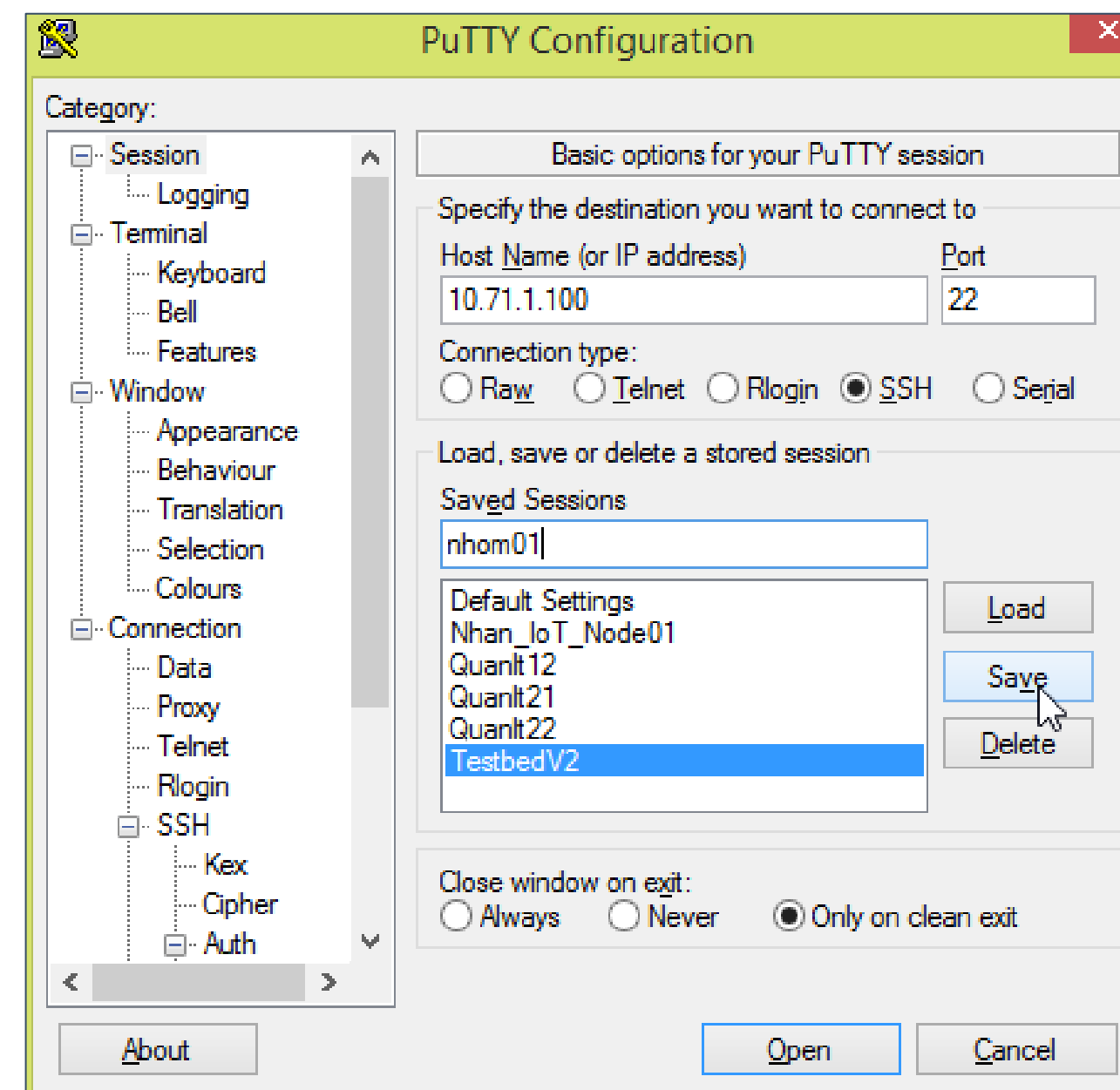
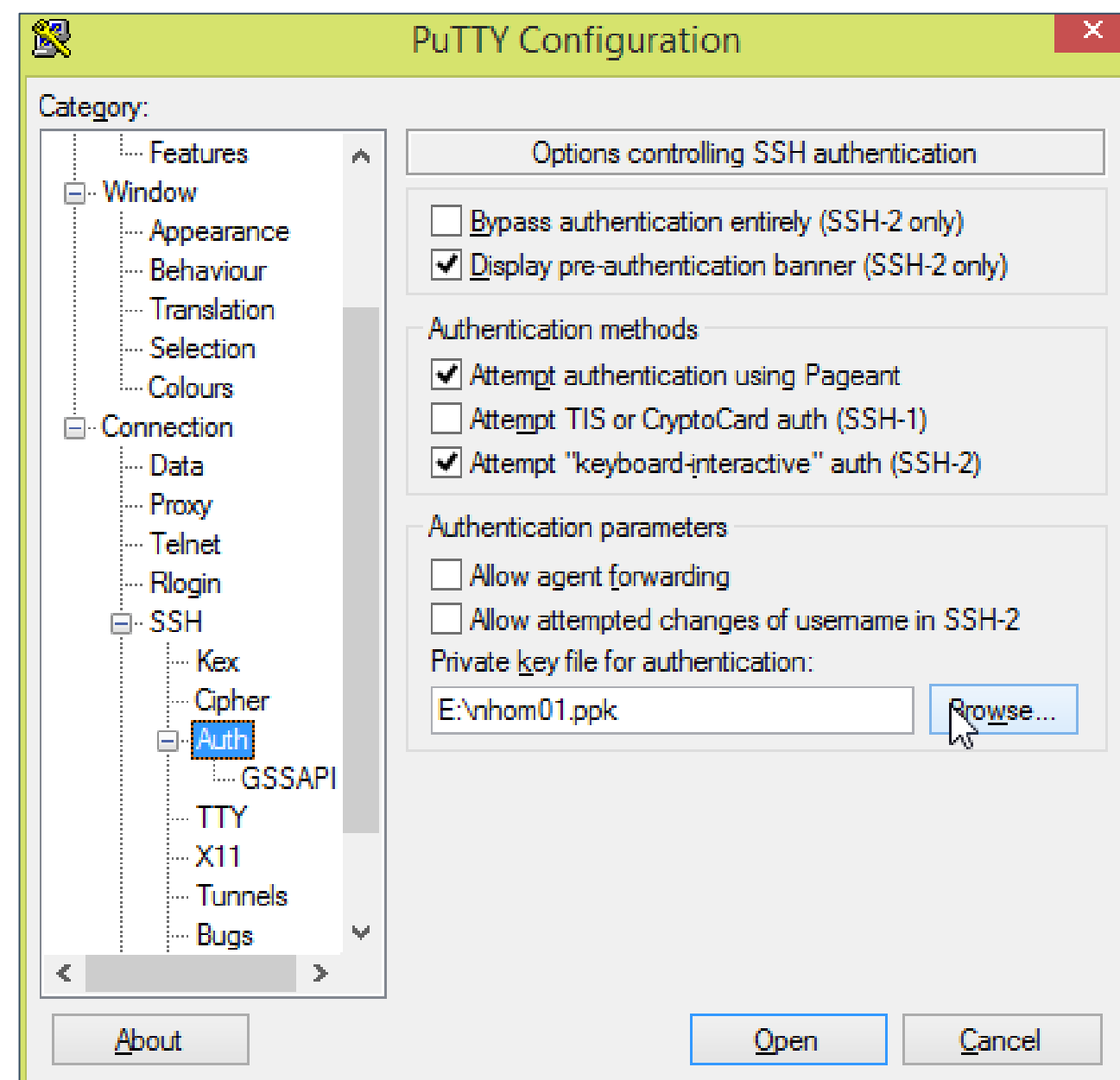
vào máy ảo

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Truy cập mạng
Private

Tạo Private Key

Kết nối SSH



```
login as: centos
Authenticating with public key "imported-openssh-key"
Last login: Tue Sep 12 03:54:06 2017 from 192.168.5.43
[centos@nhom01-pc3 ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 8950
    inet 10.10.10.58  netmask 255.255.255.0  broadcast 10.10.10.255
    inet6 fe80::f816:3eff:febb:bff2  prefixlen 64  scopeid 0x20<link>
    ether fa:16:3e:bb:bf:f2  txqueuelen 1000  (Ethernet)
    RX packets 266  bytes 27907 (27.2 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 188  bytes 32178 (31.4 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 0  (Local Loopback)
    RX packets 134  bytes 11552 (11.2 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 134  bytes 11552 (11.2 KiB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

[centos@nhom01-pc3 ~]$ sudo vi /etc/sysconfig/network-scripts/ifcfg-eth1
```

A screenshot of a terminal window titled "centos@nhom01-pc3:~". The terminal has a yellow title bar with standard window controls. The main area is black with white text. It shows several lines of configuration commands: DEVICE="eth1", ONBOOT="yes", TYPE="Ethernet", IPADDR=192.168.3.5, and NETMASK=255.255.255.0. A green cursor is at the end of the last line. Below these are several tilde (~) characters representing newlines. At the bottom left, it says "-- INSERT --". On the right side, there is a vertical scrollbar and a small icon at the bottom right corner.

```
[centos@nhom01-pc3 ~]$ sudo service network restart
Restarting network (via systemctl): [ OK ]
[centos@nhom01-pc3 ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 8950
    inet 10.10.10.58 netmask 255.255.255.0 broadcast 10.10.10.255
    inet6 fe80::f816:3eff:febb:bff2 prefixlen 64 scopeid 0x20<link>
    ether fa:16:3e:bb:bf:f2 txqueuelen 1000 (Ethernet)
    RX packets 689 bytes 67469 (65.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 463 bytes 68524 (66.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.3.5 netmask 255.255.255.0 broadcast 192.168.3.255
    inet6 fe80::f816:3eff:feb6:b1ba prefixlen 64 scopeid 0x20<link>
    ether fa:16:3e:b6:b1:ba txqueuelen 1000 (Ethernet)
    RX packets 10 bytes 888 (888.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10 bytes 676 (676.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 0 (Local Loopback)
    RX packets 134 bytes 11552 (11.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 134 bytes 11552 (11.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Demo trực tiếp tại lớp

CẤU HÌNH MÁY ẢO (BẰNG LỆNH)

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Metadata

You can customize your instance after it has launched using the options available here. "Customization Script" is analogous to "User Data" in other systems.

Customization Script (Modified) Script size: 186 bytes of 16.00 KB

```
#!/bin/bash
printf "DEVICE=\"%eth1\" \"\nONBOOT=\"%yes\" \"\nTYPE=\"%Ethernet\" \"\nIPADDR=192.168.3.6
\nNETMASK=255.255.255.0" > /etc/sysconfig/network-scripts/ifcfg-eth1
service network restart
```

Load script from a file

Lab01_initscripts.txt

Disk Partition

Automatic

☐ Configuration Drive

```
centos@nhom01-pc31:~
login as: centos
Authenticating with public key "imported-openssh-key"
[centos@nhom01-pc31 ~]$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 8950
    inet 10.10.10.59  netmask 255.255.255.0  broadcast 10.10.10.255
    inet6 fe80::f816:3eff:feac:2cf6  prefixlen 64  scopeid 0x20<link>
    ether fa:16:3e:ac:2c:f6  txqueuelen 1000  (Ethernet)
    RX packets 57  bytes 7707 (7.5 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 61  bytes 9450 (9.2 KiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

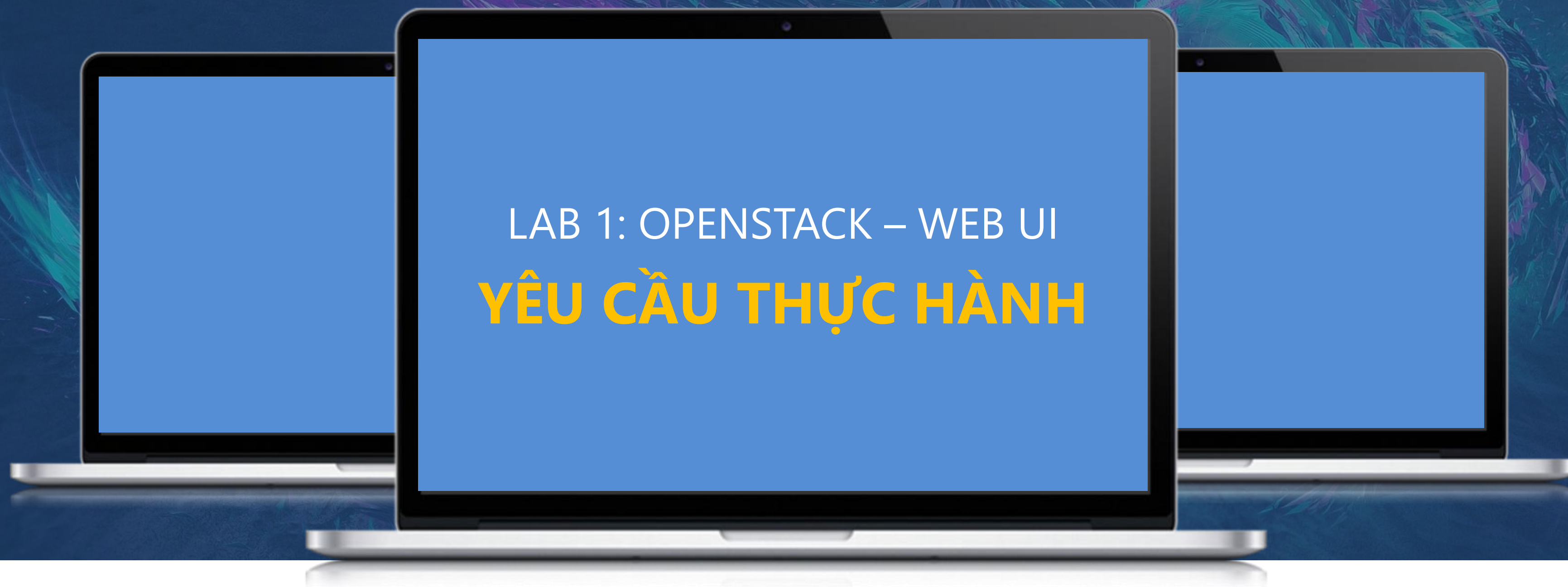
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.3.6  netmask 255.255.255.0  broadcast 192.168.3.255
    inet6 fe80::f816:3eff:fef7:669e  prefixlen 64  scopeid 0x20<link>
    ether fa:16:3e:f7:66:9e  txqueuelen 1000  (Ethernet)
    RX packets 6  bytes 480 (480.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 11  bytes 746 (746.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 0  (Local Loopback)
    RX packets 134  bytes 11552 (11.2 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 134  bytes 11552 (11.2 KiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[centos@nhom01-pc31 ~]$
```

Demo trực tiếp tại lớp

CẤU HÌNH MÁY ẢO (BẰNG INITSCRIPTS)



Yêu cầu



Hướng dẫn



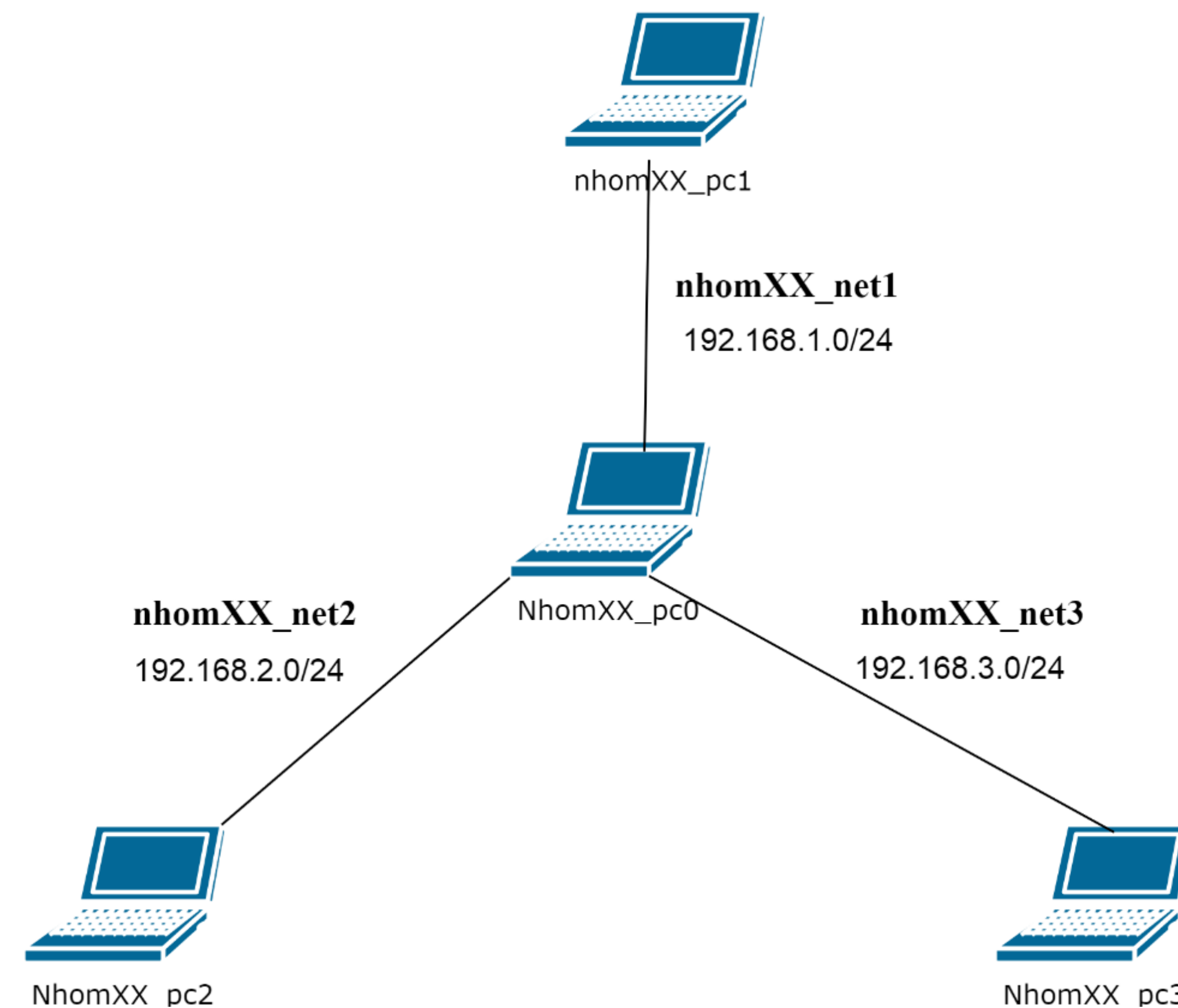
Báo cáo

YÊU CẦU

nội dung thực hành

32

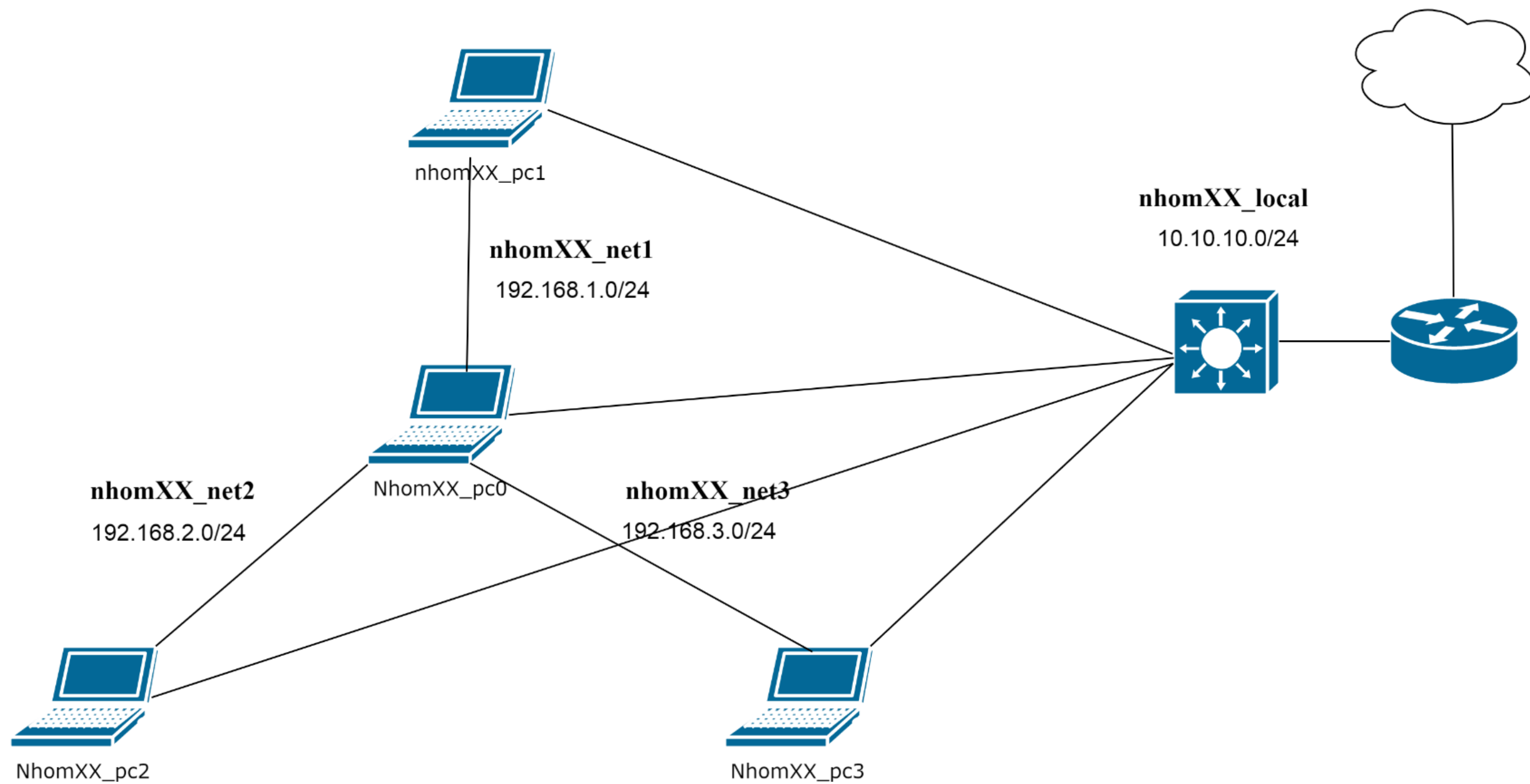
Triển khai, cấu hình hệ thống gồm 4 máy tính kết nối theo kiến trúc mạng hình sau:



YÊU CẦU

nội dung thực hành

Thực tế phải triển khai:



HƯỚNG DẪN

10 bước thực hiện

B1. Tạo network nhomXX_local

B2. Tạo router nhomXX_R kết nối network nhomXX_local

B3. Tạo các networks nhomXX_net01, nhomXX_net02, nhomXX_net03

B4. Tạo keypair: nhomXX

B5. Tạo các instances: nhomXX_pc0, nhomXX_pc1, nhomXX_pc2, nhomXX_pc3. Gắn Floating IP

B6. Add rule mở port 22, dùng putty SSH vào instances

B8. Cấu hình IP, Quagga (tham khảo hướng dẫn)

B9. Kiểm tra (ping):

- + OK: chuyển xuống viết báo cáo
- + Failed: quay lại B8.

B10. Viết báo cáo

CÂU LỆNH CẤU HÌNH

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dùng để tham khảo thực hiện bài Lab

Cấu hình IPs

```
sudo vi /etc/sysconfig/network-  
scripts/ifcfg-eth1
```

```
DEVICE="eth1"  
ONBOOT="yes"  
TYPE="Ethernet"  
IPADDR=192.168.1.5  
NETMASK=255.255.255.0
```

```
sudo service network restart
```

Cài đặt Quagga

```
sudo yum install -y quagga
```

```
sudo chmod 777 /etc/quagga
```

```
cd /etc/quagga
```

```
sudo vi zebra.conf
```

```
password centos
```

```
sudo service zebra start
```

```
sudo chkconfig zebra on
```

Cấu hình Zebra

```
sudo yum install -y telnet
```

```
telnet localhost zebra
```

```
Enable  
conf t  
interface eth1  
ipv6 nd suppress-ra  
exit  
ip forwarding  
#Do wr  
exit  
show run (kiểm tra cấu hình)  
exit (thoát khỏi telnet zebra)
```

```
sudo service zebra restart
```


CÂU LỆNH CẤU HÌNH

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dùng để tham khảo thực hiện bài Lab

Cấu hình Ripd

```
sudo cp /usr/share/doc/quagga-0.99.22.4/ripd.conf.sample /etc/quagga/ripd.conf
```

```
sudo service ripd start  
sudo chkconfig ripd on  
telnet localhost ripd (default password: zebra)
```

```
enable  
conf t  
router rip  
version 2  
network eth1  
exit  
do wr (bỏ qua vì k có quyền root)  
exit  
show run (kiểm tra kỹ cấu hình)  
exit (thoát khỏi telnet)
```

Bật IP_FORWARDING

```
sudo vi /etc/sysctl.conf
```

```
net.ipv4.ip_forward=1
```

```
sudo service network restart
```

KIỂM TRA ROUTING TABLE

```
ip route
```

```
centos@nhom01-pc1:/etc/quagga  
[centos@nhom01-pc1 quagga]$ ip route  
default via 10.10.10.1 dev eth0 proto static  
10.10.10.0/24 dev eth0 proto kernel scope link src 10.10.10.61  
169.254.169.254 via 10.10.10.1 dev eth0 proto static  
192.168.1.0/24 dev eth1 proto kernel scope link src 192.168.1.6  
192.168.2.0/24 via 192.168.1.5 dev eth1 proto zebra metric 2  
192.168.3.0/24 via 192.168.1.5 dev eth1 proto zebra metric 2  
[centos@nhom01-pc1 quagga]$
```

```
centos@nhom01-pc2:/etc/quagga  
[centos@nhom01-pc2 quagga]$ ip route  
default via 10.10.10.1 dev eth0 proto static  
10.10.10.0/24 dev eth0 proto kernel scope link src 10.10.10.62  
169.254.169.254 via 10.10.10.1 dev eth0 proto static  
192.168.1.0/24 via 192.168.2.3 dev eth1 proto zebra metric 2  
192.168.2.0/24 dev eth1 proto kernel scope link src 192.168.2.6  
192.168.3.0/24 via 192.168.2.5 dev eth1 proto zebra metric 2  
[centos@nhom01-pc2 quagga]$
```

```
centos@nhom01-pc3:/etc/quagga  
[centos@nhom01-pc3 quagga]$ ip route  
default via 10.10.10.1 dev eth0 proto static  
10.10.10.0/24 dev eth0 proto kernel scope link src 10.10.10.63  
169.254.169.254 via 10.10.10.1 dev eth0 proto static  
192.168.1.0/24 via 192.168.3.3 dev eth1 proto zebra metric 2  
192.168.2.0/24 via 192.168.3.7 dev eth1 proto zebra metric 2  
192.168.3.0/24 dev eth1 proto kernel scope link src 192.168.3.8  
[centos@nhom01-pc3 quagga]$
```

TROUBLESHOOT

ping vẫn chưa được?

Vì mặc định Openstack chặn forward gói tin qua các network ports!

The screenshot shows the OpenStack Dashboard with the 'Networks / net1 / Ports' page. The left sidebar contains navigation links for Project, Compute, Network, Routers, Orchestration, Object Store, and Identity. The main content area displays details for a specific port (ID: 2e6e00b5-c051-415d-9bb5-80c68a63b658) associated with the network 'net1'. The details are organized into sections: General information (Name, ID, Network Name, Network ID, Project ID, MAC Address, Status, Admin State, DNS Name), DNS Assignment (Hostname, IP Address, FQDN), Fixed IPs (IP Address, Subnet ID), Attached Device (Device Owner, Device ID), Binding, and VNIC Type.

Property	Value
Name	None
ID	2e6e00b5-c051-415d-9bb5-80c68a63b658
Network Name	net1
Network ID	26fb823c-6709-41e1-aff6-69cd8a3c2b70
Project ID	0a837ff311b24eaaa2bc933c5b6f2f65
MAC Address	fa:16:3e:31:71:6d
Status	Active
Admin State	UP
DNS Name	None
DNS Assignment	
Hostname	host-192-168-1-5
IP Address	192.168.1.5
FQDN	host-192-168-1-5.openstacklocal.
Fixed IPs	
IP Address	192.168.1.5
Subnet ID	f9d67be0-40dd-48cc-9b3a-8a71d57f0fee
Attached Device	
Device Owner	compute:nova
Device ID	9243b464-3f25-4f99-8f70-815299164b5c
Binding	
VNIC Type	Normal

The screenshot shows a 'Request' editor interface. The URL bar contains 'https://mitaka.uit.edu.vn:5000/v3/auth/tokens'. The method is set to 'POST'. The content type is 'application/json'. The 'Raw headers' tab is selected, showing 'Content-Type: application/json'. The 'Raw payload' tab is also selected, displaying a JSON body for authentication. The body includes fields for 'auth', 'identity', 'methods', 'password', and 'user'. The 'user' field contains a name and domain, and the 'password' field is masked. A 'SEND' button is visible at the bottom right.

```
{
  "auth": {
    "identity": {
      "methods": [
        "password"
      ],
      "password": {
        "user": {
          "name": "[redacted]",
          "domain": {
            "name": "Default"
          }
        },
        "password": "[redacted]"
      }
    }
  }
}
```

TROUBLESHOOT

ping vẫn chưa được?

201 Created 62.00 ms DETAILS ^

Response headers 8 Request headers 2 Redirects 0 Timings

Connection: close
Content-Length: 9488
Content-Type: application/json
Date: Tue, 12 Sep 2017 10:01:01 GMT
Server: Apache
Vary: X-Auth-Token
X-Openstack-Request-Id: req-d7624cfb-5144-4b44-be23-9b895e838d7d
X-Subject-Token: gAAAAABZt7BdRYvW4-H1qBDUwmKBoyZnH4FA_WfXgORn_VhV80PheIXAh0jdH7vHvYkLM7oP
t5IS5UH8Rp6-1dgBmNvYD2mSNZ4yHDEGOuWJ5aS-WkhkaxEG16wFA8hZoSQZ_8AbbYVhNcS2Y1haL4IX7bbW0Ao0v
-1q39_ByZxI48gBb01YV

Copy Ctrl+C

Raw JSON

Request Use XHR

> https://mitaka.uit.edu.vn:9696/v2.0/ports/2e6e00b5-c051-415d-9bb5-80c68a63b658

GET POST PUT DELETE PATCH Other methods

application/json

Raw headers Headers form Headers sets Variables

Content-Type: application/json
X-Auth-Token: gAAAAABZt7BdRYvW4-H1qBDUwmKBoyZnH4FA_WfXgORn_VhV80PheIXAh0jdH7vHvYkLM7oP

Raw payload Data form Files

{"port":{"security_groups":null, "port_security_enabled":false}}

229 bytes

SEND

200 OK 300.00 ms DETAILS ^

Response headers 5 Request headers 3 Redirects 0 Timings

Connection: close
Content-Length: 809
Content-Type: application/json; charset=UTF-8
Date: Tue, 12 Sep 2017 10:04:09 GMT
X-Openstack-Request-Id: req-430638bd-1bd4-4a03-b918-09a10c008227

```
centos@nhom01-pc1:/etc/quagga
[centos@nhom01-pc1 quagga]$ ping 192.168.2.6
PING 192.168.2.6 (192.168.2.6) 56(84) bytes of data.
64 bytes from 192.168.2.6: icmp_seq=1 ttl=63 time=1.10 ms
64 bytes from 192.168.2.6: icmp_seq=2 ttl=63 time=1.13 ms
64 bytes from 192.168.2.6: icmp_seq=3 ttl=63 time=4.46 ms
64 bytes from 192.168.2.6: icmp_seq=4 ttl=63 time=1.19 ms
64 bytes from 192.168.2.6: icmp_seq=5 ttl=63 time=1.16 ms
^C
--- 192.168.2.6 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 1.100/1.814/4.468/1.327 ms
[centos@nhom01-pc1 quagga]$ ping 192.168.3.8
PING 192.168.3.8 (192.168.3.8) 56(84) bytes of data.
64 bytes from 192.168.3.8: icmp_seq=1 ttl=63 time=2.23 ms
64 bytes from 192.168.3.8: icmp_seq=2 ttl=63 time=0.977 ms
64 bytes from 192.168.3.8: icmp_seq=3 ttl=63 time=1.00 ms
64 bytes from 192.168.3.8: icmp_seq=4 ttl=63 time=0.996 ms
^C
--- 192.168.3.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 0.977/1.302/2.234/0.538 ms
[centos@nhom01-pc1 quagga]$
```

```
centos@nhom01-pc2:/etc/quagga
[centos@nhom01-pc2 quagga]$ ping 192.168.1.6
PING 192.168.1.6 (192.168.1.6) 56(84) bytes of data.
64 bytes from 192.168.1.6: icmp_seq=1 ttl=63 time=12.2 ms
64 bytes from 192.168.1.6: icmp_seq=2 ttl=63 time=1.23 ms
64 bytes from 192.168.1.6: icmp_seq=3 ttl=63 time=1.10 ms
64 bytes from 192.168.1.6: icmp_seq=4 ttl=63 time=1.10 ms
^C
--- 192.168.1.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 1.106/3.921/12.240/4.803 ms
[centos@nhom01-pc2 quagga]$ ping 192.168.3.8
PING 192.168.3.8 (192.168.3.8) 56(84) bytes of data.
64 bytes from 192.168.3.8: icmp_seq=1 ttl=63 time=11.0 ms
64 bytes from 192.168.3.8: icmp_seq=2 ttl=63 time=0.914 ms
64 bytes from 192.168.3.8: icmp_seq=3 ttl=63 time=0.868 ms
64 bytes from 192.168.3.8: icmp_seq=4 ttl=63 time=0.798 ms
^C
--- 192.168.3.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 0.798/3.405/11.043/4.410 ms
[centos@nhom01-pc2 quagga]$
```

```
centos@nhom01-pc3:/etc/quagga
[centos@nhom01-pc3 quagga]$ ping 192.168.1.6
PING 192.168.1.6 (192.168.1.6) 56(84) bytes of data.
64 bytes from 192.168.1.6: icmp_seq=1 ttl=63 time=2.38 ms
64 bytes from 192.168.1.6: icmp_seq=2 ttl=63 time=0.986 ms
64 bytes from 192.168.1.6: icmp_seq=3 ttl=63 time=0.964 ms
64 bytes from 192.168.1.6: icmp_seq=4 ttl=63 time=0.963 ms
^C
--- 192.168.1.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 0.963/1.324/2.383/0.611 ms
[centos@nhom01-pc3 quagga]$ ping 192.168.2.6
PING 192.168.2.6 (192.168.2.6) 56(84) bytes of data.
64 bytes from 192.168.2.6: icmp_seq=1 ttl=63 time=5.69 ms
64 bytes from 192.168.2.6: icmp_seq=2 ttl=63 time=0.923 ms
64 bytes from 192.168.2.6: icmp_seq=3 ttl=63 time=0.888 ms
64 bytes from 192.168.2.6: icmp_seq=4 ttl=63 time=0.883 ms
^C
--- 192.168.2.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 0.883/2.096/5.692/2.076 ms
[centos@nhom01-pc3 quagga]$
```


BÁO CÁO

cần gì?

Quy trình thực hiện

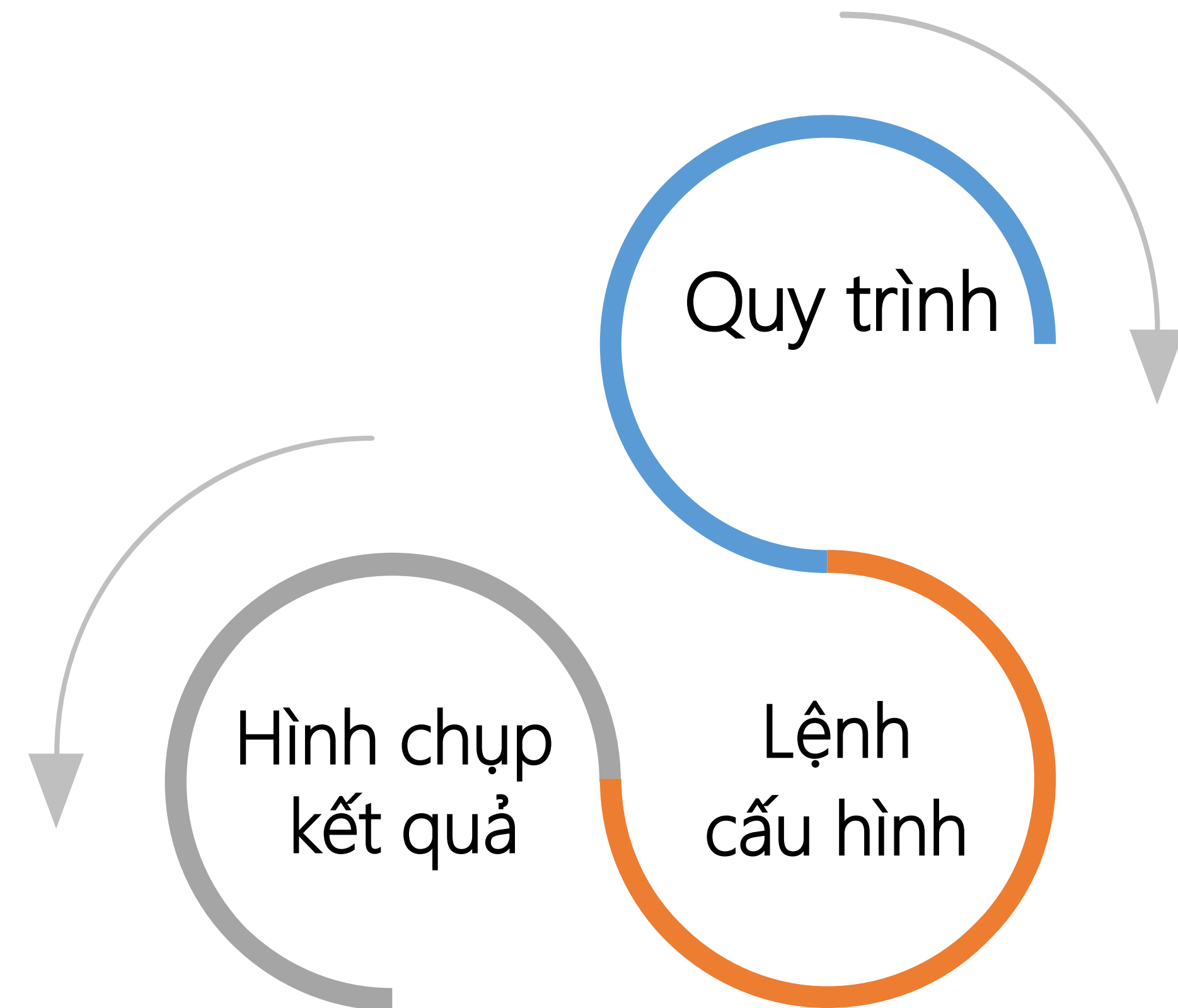
Hình chụp quy trình các bước thực hiện

Lệnh cấu hình

Nội dung lệnh cấu hình cho từng máy

Hình chụp kết quả

Hình chụp kết quả ping giữa các máy





Now It's your turn!